Panda

AI-Powered Fresh Food Demand & Inventory Optimization



PANDA Team · EWA · InnovatiCS · Group 10

SAVOLA

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AGENDA

What we will cover today

FOUR STEPS

1. **Business understanding** — the problem, the targets and the questions
2. **Data understanding** — what the data tells us about stock outs, shelf life and waste
3. **Data preparation** — how we cleaned it and kept the test fair
4. **Modeling & recommendations** — eight approaches, how they decide, and what Panda should do next

Industry-Specific Problem Statement

PROBLEM STATEMENT

Panda Retail's supply-chain roles focus on store/DC replenishment, product availability, overstock, and aged and near-expiry inventory. This project will forecast SKU-store demand and optimize replenishment for fresh and short-shelf-life products, where excess stock creates markdown/expiry waste while insufficient stock causes lost sales.
For the academic dataset, assume a fresh-food forecast MAPE of 25–30%, an 8–12% stockout rate for high-velocity fresh SKUs during promotion/peak periods, and 5–8% of fresh inventory units entering markdown, expiry, or disposal status.

BUSINESS OBJECTIVES

1. **Forecast store-level demand** (units sold) for fresh products with Saudi calendar and promotion effects, reducing forecast error (MAPE) from 25–30% to below 15% for high-velocity products

2. **Predict stockout and near-expiry risk**, achieving at least 85% recall for near-expiry risk and cutting promo stockouts on fresh products by 25–30% (from a baseline of 8–12%)

3. **Optimize DC/store replenishment and allocation** under shelf-life constraints, improving on-shelf availability by 3–5 percentage points

4. **Quantify lost sales, markdown, expiry waste and working-capital improvement**, cutting fresh stock going to markdown or expiry by 20–25% (from a baseline of 5–8%)

DATA AT A GLANCE

Two years of Panda fresh-food data: big enough and clean enough to build on

1M
inventory & sales records

730
days · Jan 2025 – Dec 2026

1,000
stores · 30 distribution centres

150K
products (SKUs) · 12,000 suppliers

83.8%
records fully complete

0
duplicate rows or IDs

0
invalid dates or flags

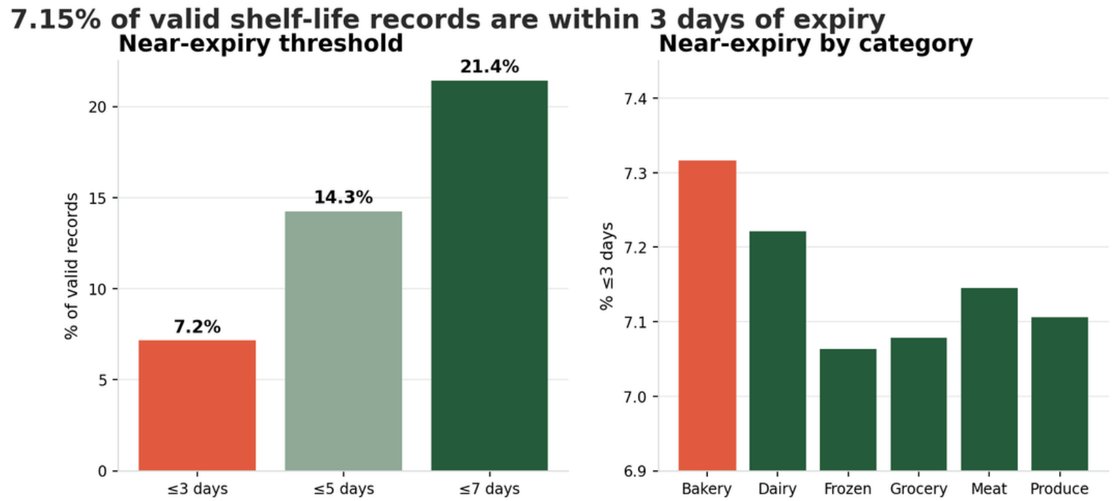
3 of 26
columns with gaps (4–8%)

SO WHAT

The data covers every store and two full years, so the patterns on the next slide are reliable. The few gaps are small and are fixed in Phase 3 before any model is built.

DESCRIPTIVE QUESTION · DQ4

Remaining Shelf Life and Near-Expiry Share



7.15%

of valid shelf-life records are within 3 days of expiry

67,950

of 949,813 valid shelf-life records

≤ 3 days

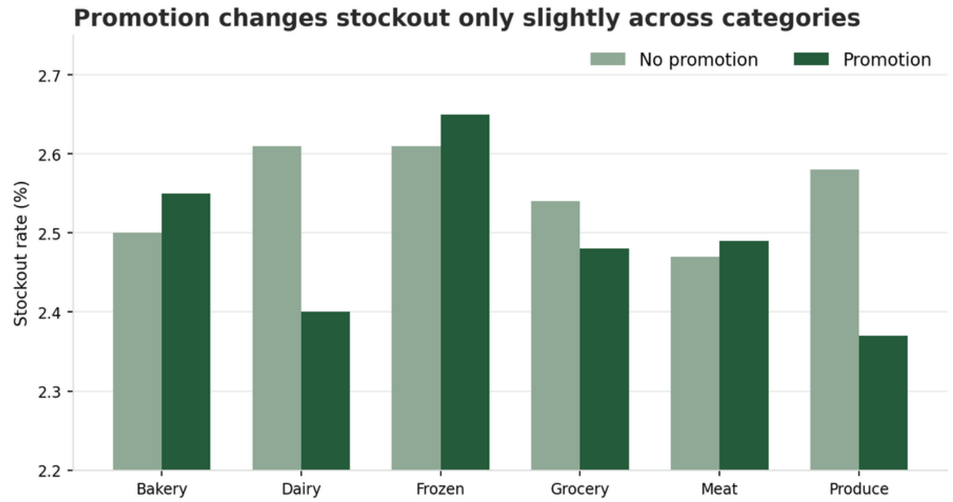
exploratory threshold, to be confirmed with the business team in Phase 3

QUESTION

What is the distribution of remaining shelf life across product categories and stores, and what proportion of inventory records is near expiry?

DESCRIPTIVE QUESTION · DQ2

Stockout Rates: Promotion vs Non-Promotion



2.55%

pooled stockout rate without promotion

2.49%

pooled stockout rate during promotion

–0.06 pts

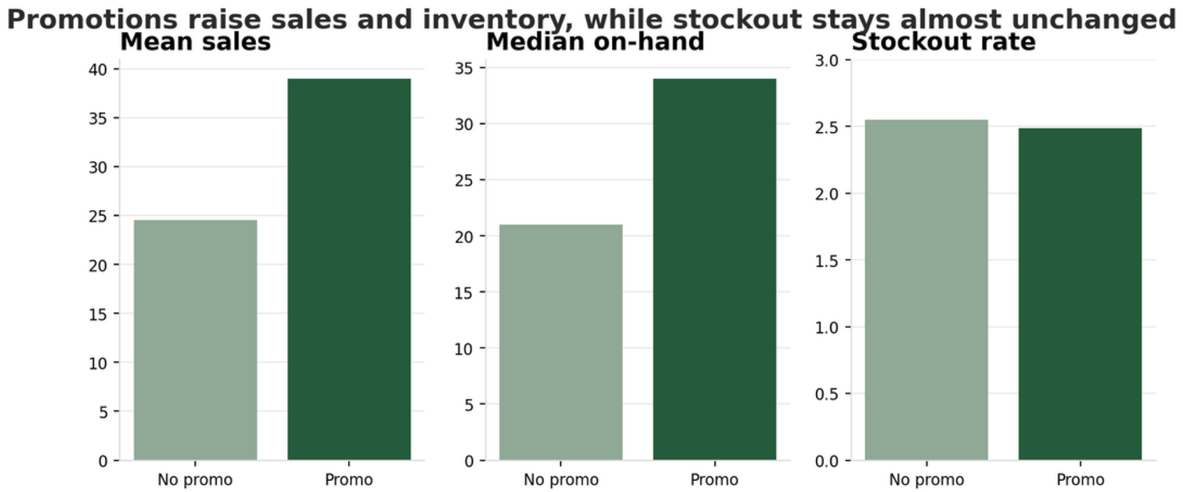
promotions raise mean sales, but not the pooled stockout rate

QUESTION

What differences exist in stockout rates between promotional and non-promotional periods across stores and product categories?

DIAGNOSTIC QUESTION · GQ3

Why Stockout Differs Between Promotion and Non-Promotion



24.55 → 38.98

Mean Sale: no promotion vs promotion (+59%)

2.55% vs 2.49%

stockout rate without vs with promotion: keep stable

QUESTION

Why do stockout rates differ between promotional and non-promotional periods, considering sales volumes, on-hand inventory, and supplier lead times?

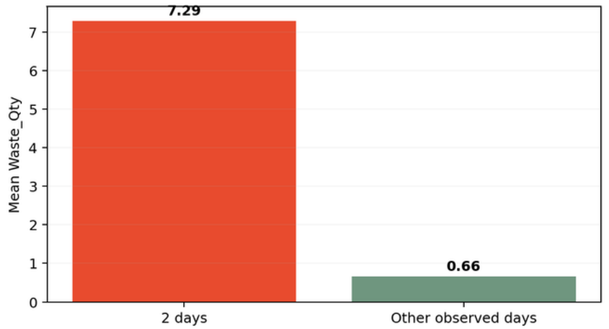
WHAT THE DATA TELLS US

Three findings Panda can act on

1 · WASTE

27.8%

of all waste comes from the **3.6%** of stock with 2 days left



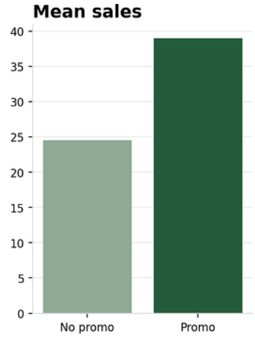
Mean waste per record: 7.29 at 2 days vs 0.66

So what: mark down or transfer 2–3 days before expiry

2 · DEMAND

+59%

more units sold on promotion, and **≈3×** more in the busiest stores



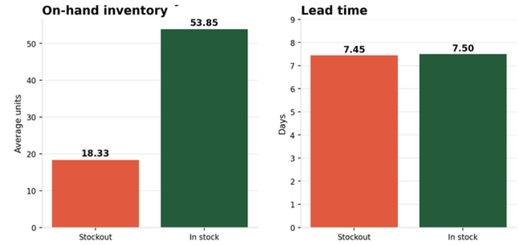
Mean sales: 24.55 → 38.98 on promotion

So what: forecast with the promotion calendar and store traffic

3 · STOCKOUTS

−66%

less stock on hand when shelves run short; supplier lead time is the **same**



During promotions: on hand 18.33 vs 53.85

So what: fix how much is replenished, not which supplier

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PHASE 3 · DATA PREPARATION

Data Preparation & Modeling Readiness

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From 1,000,000 raw records to 871,314 we can trust

128,686

Records from the future

Dated after 27 Sep 2026, so no real sales exist yet. Excluded.

9,825

Inflated sales records

Units sold were far higher than the money collected implies. Corrected.

964

Hidden stockouts

Shelf stock was zero but no stockout was recorded. Relabelled.

Why it matters Left uncleaned, the model would over-forecast demand, we would over-order, and fresh food waste would go up.

Source: PANDA.csv (1,000,000 records × 26 columns) · Phase 3 notebook · synthetic academic dataset

HOW WE FIXED IT

We fixed three data problems without throwing away real business signals

Missing values

8% → 0%

BEFORE AFTER

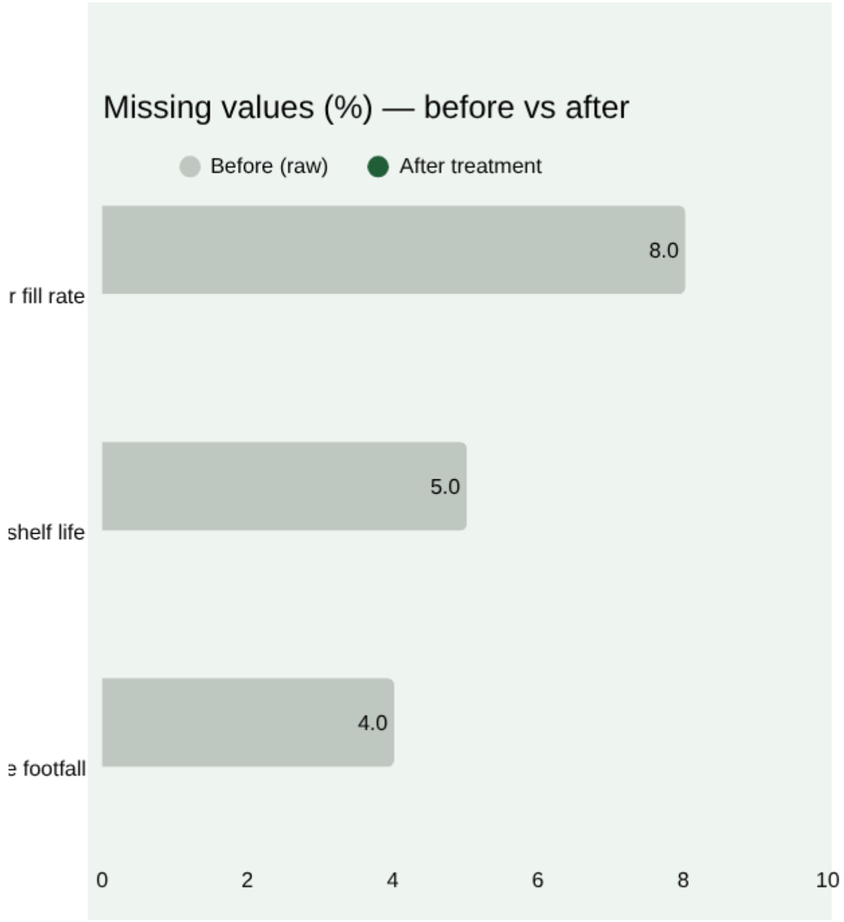
of supplier fill-rate values were empty

WHAT WE DID

Filled with the typical value for the same supplier, plus a “was missing” flag.

WHY

Deleting rows would lose data in 3 columns, and a missing value can itself signal a supply problem.



HOW WE FIXED IT

We fixed three data problems without throwing away real business signals

Extreme values

4.04 → 0.02

BEFORE AFTER

skew of weekly sales (how lopsided it is)

WHAT WE DID

Kept real peaks. Capped predictors at the 99.9th percentile and log-transformed sales.

WHY

Promotions and big stores create real demand. Deleting them would hide the busiest days.

Text categories

7 → 6

BEFORE AFTER

category labels (“ produce ” vs Produce)

WHAT WE DID

Cleaned labels and one-hot encoded small fields. IDs are encoded by frequency instead.

WHY

A product spelled two ways counts twice. One-hot on 149,827 SKU IDs would explode the table.

HOW WE WILL ANALYSE IT

Analytical Approach

1 · FORECASTING

< 15%

forecast error, from 25–30%

PREDICTS

Weekly units sold per store and category

Model SARIMA / Prophet

2 · REGRESSION

−20–25%

fresh-food waste

PREDICTS

Waste quantity per batch

Model Random Forest / XGBoost

3 · CLASSIFICATION

≥ 85%

of stockout and expiry risks caught

PREDICTS

Stockout and near-expiry: yes / no

Model Logistic regression / XGBoost

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PHASE 4 · MODELING

Modeling & Business Results

From data to daily decisions for Panda fresh food

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PHASE 4 · SUMMARY

Overview: from Panda's data to daily decisions

WHAT WE BUILT

Eight machine-learning approaches that turn Panda's daily records into one morning routine.

THE DATA SHOW US

Stock outs come from **low shelves**, not slow suppliers: stock on hand drives 83% of the risk.

Waste comes from **high stock close to expiry**: those records waste 2+ units in 66% of cases.

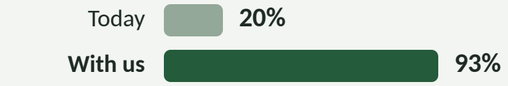
BUILT BY THE PANDA TEAM

Sara Alshamrani (Team Lead) · Afnan Aljohani · Khalid Alhrabi · Shahad Marwi · Rahaf Alhadhrami

TODAY → WITH OUR MODELS

1

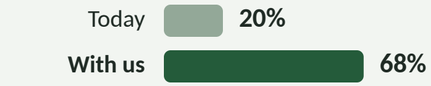
Stock out risk prediction



of stock outs found checking 1 in 5 records: **4.6× more** than today

2

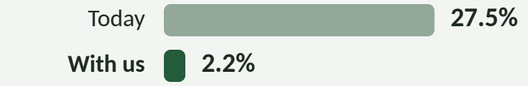
Waste quantity prediction



of wasted units on one daily list; **SAR 23K** is wasted every day

3

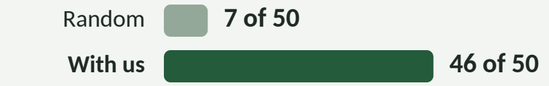
Daily unit demand forecast



forecast error, against 25–30% today: the **15% target is met**

4

Patterns & next best action



right picks each morning; unusual records waste **11× more**

Today = how Panda decides without a model (random checks, Phase 1 error assumption) · tested once on 8 weeks never seen · full summary matrix in Appendix D0

PHASE 4 · SOLUTION 1 · PREDICTION

Stock out Risk and Waste Quantity Prediction

STOCK OUT RISK PREDICTION

Same checks, 4.6× more stock outs found.

Share of stock outs found when 1 in 5 records is checked



35 stock outs found a day, against 8 today

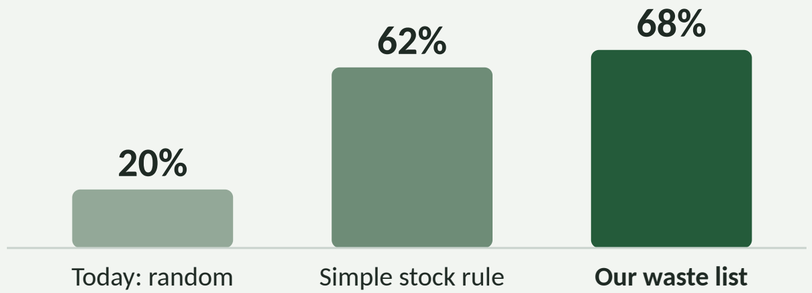
7.8× lift in the riskiest 1 in 10 records

Accuracy 96.6% **F1 0.43** best of 9 models · tied with Khalid's LightGBM (96.6%, F1 0.44)

WASTE QUANTITY PREDICTION

Two-thirds of the waste on one list.

Share of wasted units in the 1 in 5 records flagged



SAR 23K of fresh food wasted every day (1,218 units)

48% smaller error than using the average

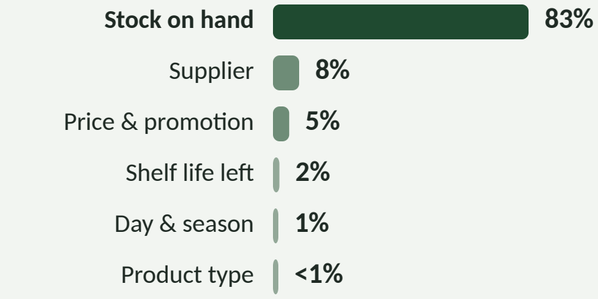
THE DATA SHOW US

Stock outs come from low shelves and waste from high stock close to expiry, so one morning list catches both. The Phase 1 target of 85% is met (93%).

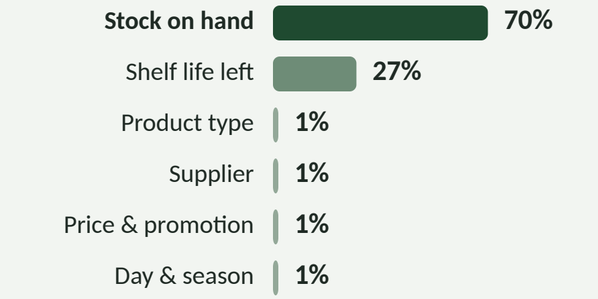
Baselines: random checks (today), a simple low-stock / stock-per-shelf-day rule · 95% range 92%–94% and 67%–68% · engine: XGBoost for both lists · Appendix D1–D2, F1–F2

Inside the Black Box: How the Data Drive Each Decision

WEIGHT OF EACH SIGNAL · STOCK OUT



WEIGHT OF EACH SIGNAL · WASTE



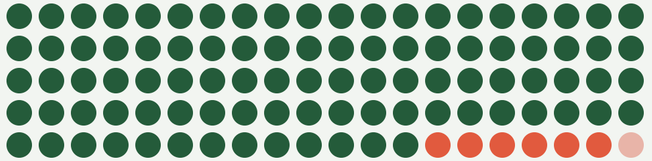
⚠ THE RISKIEST SIGNALS

Under 9 units on the shelf: the stock out risk turns sharply up.

Over 25 units with **3 days or less** of shelf life: waste climbs.

Supplier is the next signal for stock outs (8%).

RIGHT OR WRONG · OUT OF 100 STOCK OUTS



93 found: 100% of them had under 20 units on the shelf.

7 missed: some stock was still on the shelf (about 12 units), so the risk looked moderate; a sudden rush is not in the data.

found · missed · missed with 20+ units

THE DATA SHOW US

The model thinks like a good store manager: stock on hand carries 83% of the stock out decision; stock and shelf life carry 97% of the waste decision.

Colours: dark green = strongest signal; green = found, red = missed · method: SHAP impact on 4,000 test records, association not proof of cause · Appendix D3–D4, F4–F5

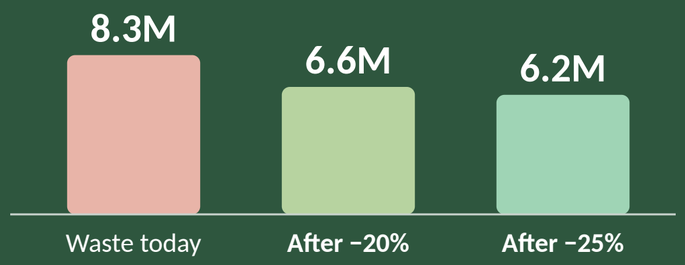
PHASE 4 · RECOMMENDATION · BUSINESS CASE

Why Panda Should Invest: One Morning Routine

THE BUSINESS CASE

SAR 1.7M–2.1M a year saved if the pilot confirms the Phase 1 waste target (−20–25%).

Fresh-food waste per year in this dataset, SAR millions



4.6× more stock outs caught before customers notice

< 1 s to score a full day, on data Panda already has

THE ASK · NEXT STEPS

1. **Weeks 1–3:** pilot in a few stores. Pass = waste down 20%, 85% of stock outs caught
2. **If passed:** more stores, same morning routine
3. **Then:** the whole network

Needs: no new devices · Panda's existing data · the same teams

WHO DOES WHAT · EXPECTED IMPACT · DRIVER · MONITOR

1. **Store manager** · Check the morning risk list first
   Expected: **35 stock outs found a day (today 8)**
   driver: under 9 units on the shelf · monitor: found per 100 checks

2. **Category manager** · Mark down or transfer the flagged items
   Expected: **68% of wasted units on one list**
   driver: over 25 units, ≤ 3 days of shelf life · monitor: waste in SAR

3. **Supply planner** · Set tomorrow's volume the day before
   Expected: **±3% on 7 days in 10**
   driver: recent sales, promotions · monitor: daily forecast error

4. **Data owner** · Review unusual records; record expiry at receiving
   Expected: **flagged records waste 11× more**
   driver: very high stock, little on order · monitor: % records with shelf life

LIMITS Tested on an academic dataset · the models show what goes together, not what causes it · a sudden rush of customers is not in the data, so the pilot is the real test.

Value: waste units × effective price on the 8 test weeks (SAR 23K a day, 1,370 records a day), scaled to a year; savings are a scenario for the pilot to prove, not a measured result · academic dataset

PANDA | Sales & Operations Dashboard

Category
All

Select Month
Mar

Current Mont..
3M 3M

Promotion Flag
Without Pr..
With Prom..

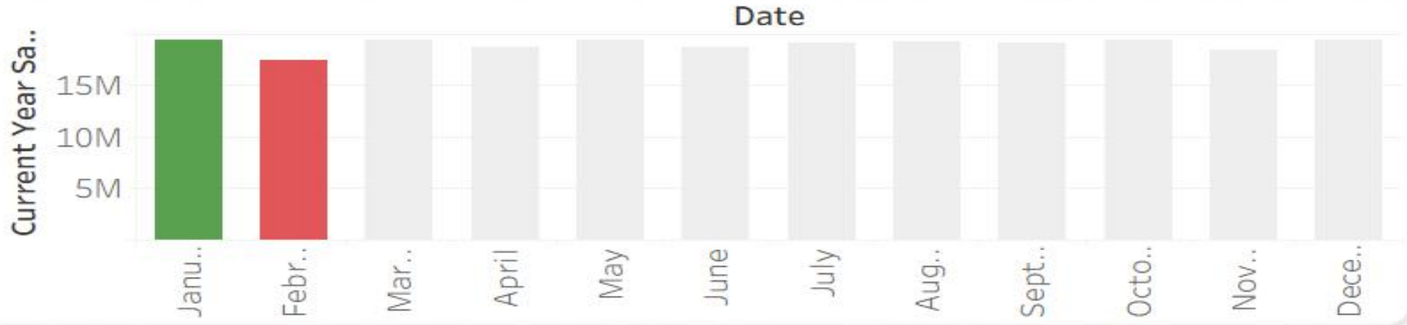
Highest Stock..
Highest
Other

Highest Lowe..
Highest
Lowest
Other

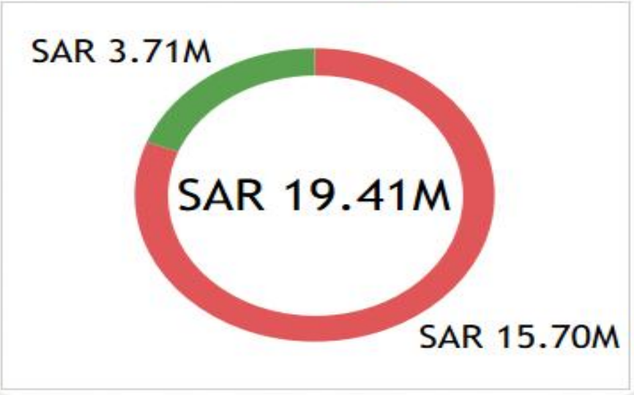
Total Sales	Waste Qty	Stockout Rate	Promotion Sales Share	Unit Sales
SAR 19.41M	37,247	2.5%	19.1%	1.14M
-0.02% vs. PY	-2.0% vs. PY	-0.03 pp vs. PY	-1.1 pp vs. PY	0.24% vs. PY

Total Sales Value | Mar

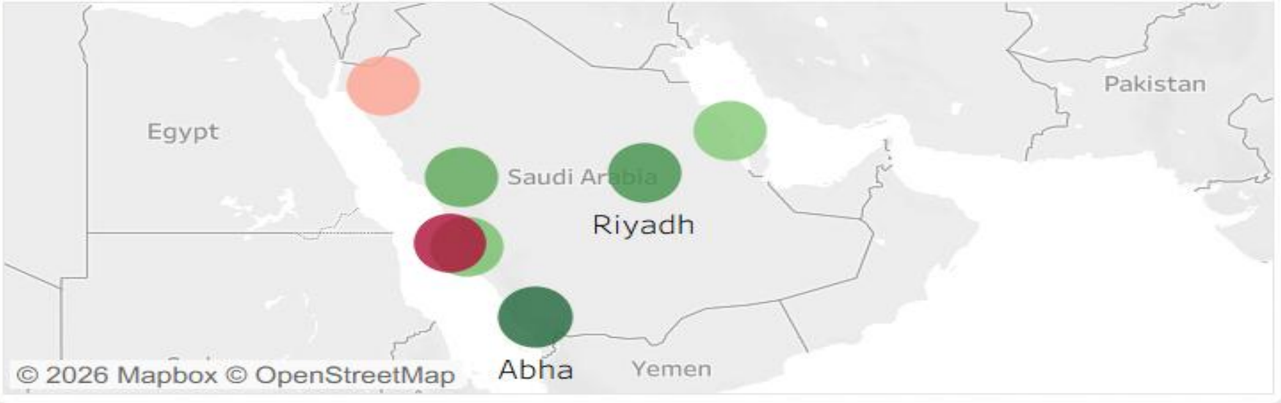
SAR 19.41M 0.0% vs. PY



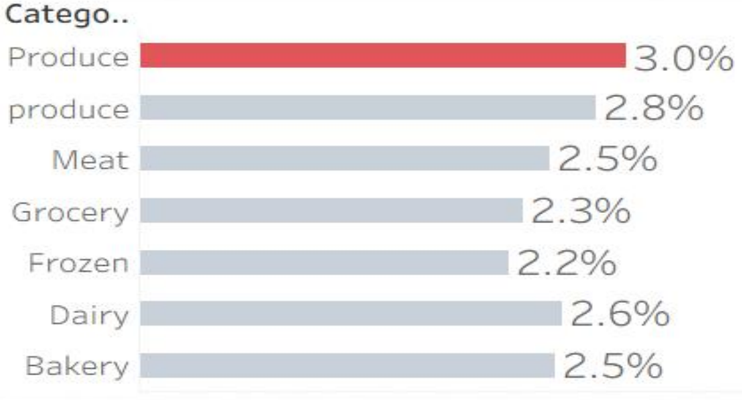
Promotion vs Regular Sales



Sales by City



Stockout Rate by Category



Panda Visit

زيارة شركة بنده

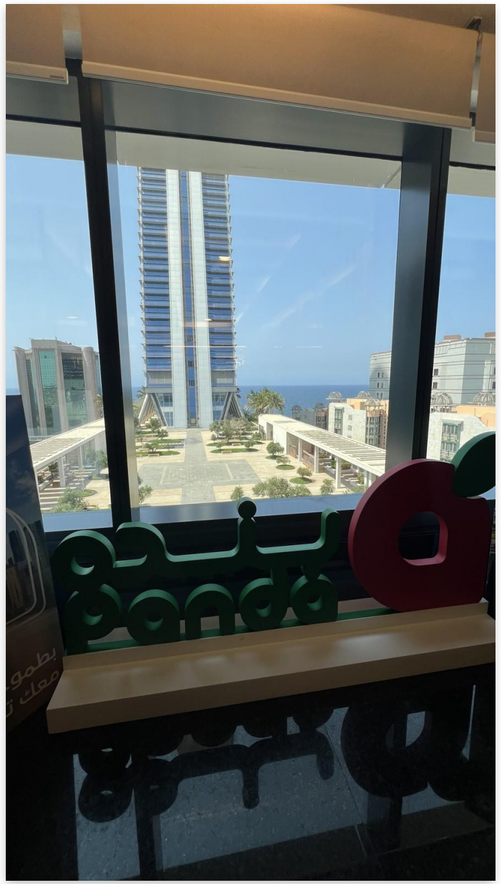
Savola Group HQ · Jeddah



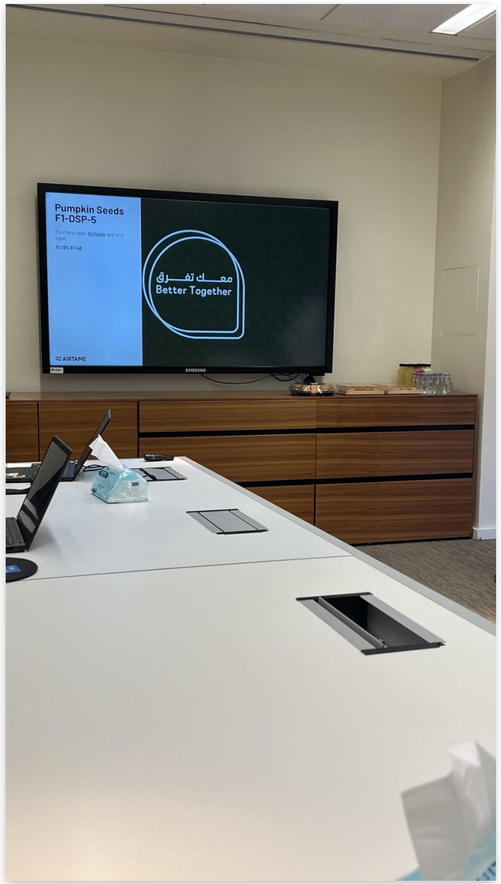




Inside the Visit









معك تفرق

Better Together

Panda offices · Ocean-view workspace · Meeting session

EXPLORE THE PROJECT

See the full Panda project online

بندة · تحسين الطلب والمخزون للأغذية الطازجة

Panda · Fresh-food demand & inventory optimization

afnanaljohani.github.io/Afnan/#deck

Click to open the interactive project site →

Scan to open on your phone

Interactive site: every phase, the models and the business case in one place

Panda

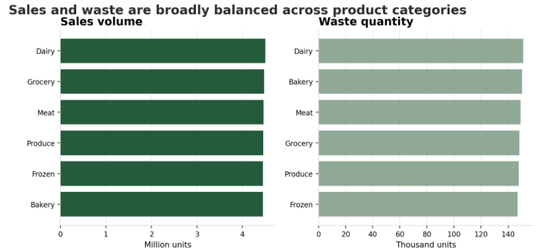
THANK YOU

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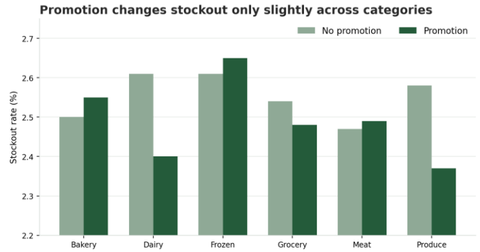
D1–D5: what happened

D1 · Sales and waste by category



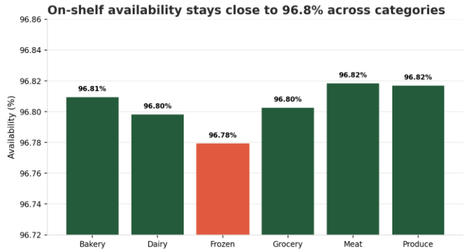
Sales and waste are similar in all 6 categories

D2 · Stockout: promotion vs not



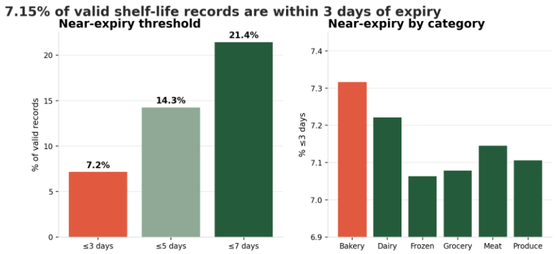
Promotion barely moves stockout (2.49% vs 2.55%)

D3 · Shelf availability



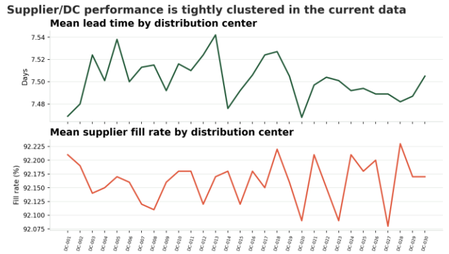
Availability ≈ 96.8% everywhere; Frozen lowest

D4 · Near expiry



7.15% of stock is within 3 days of expiry

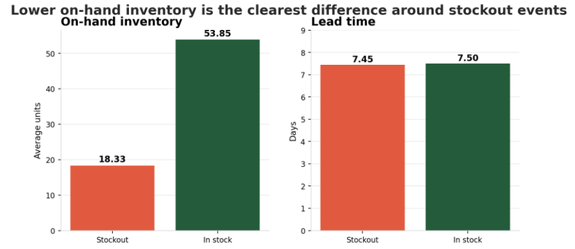
D5 · Lead time and fill rate



Lead time ≈ 7.5 days, fill rate ≈ 92% at every DC

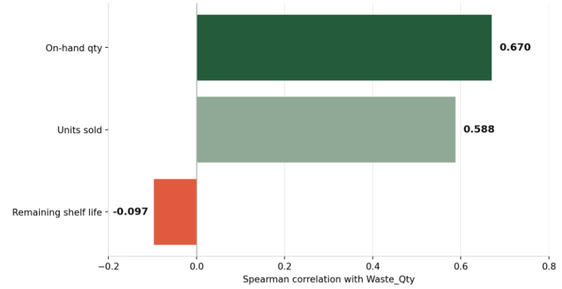
G1–G5: why it happened

G1 · Why stockouts vary



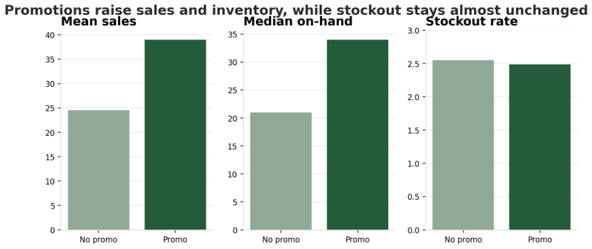
Stockouts come with low stock, not slow suppliers

G2 · Why waste varies



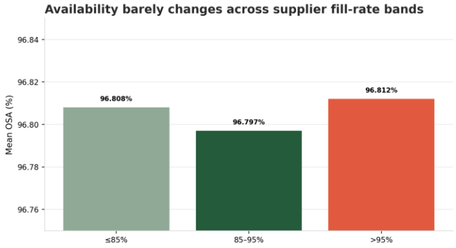
Waste moves with stock on hand (ρ 0.67)

G3 · Promotion vs stockout



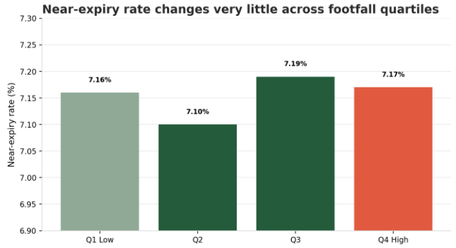
Promotions lift sales +59%; stockout unchanged

G4 · Why availability varies



Supplier fill rate does not explain availability

G5 · Why near-expiry varies



Footfall does not change near-expiry (7.1–7.2%)

APPENDIX D0 · SOLUTIONS SUMMARY MATRIX

Eight approaches, each one beats today's way of working

BUSINESS QUESTION	APPROACH · CHAMPION	TODAY (BASELINE)	WITH OUR MODELS (TEST)	ACTION
Which records will run out tomorrow?	**Stock out risk prediction** classification · XGBoost	random checks find **2 in 10**	**9 in 10** found, same checks accuracy 96.6% · F1 0.43	Morning risk list
How much will each record waste?	**Waste quantity prediction** regression · XGBoost	a random list covers **20%** of waste	**68%** of waste on one list	Mark-down list
How many units sell tomorrow?	**Daily unit demand forecast** time series · ETS	**25–30%** error (Phase 1)	**2.2%** error · ±3% on 7 of 10 days	Set tomorrow's volume
Which records are unusual?	**Finding the unusual** anomaly detection · Robust z-score	no review list	flagged records waste **11×** more	Daily review list
Which records behave alike?	**Natural groups** clustering · K-Means	every record treated the same	one group (6%) wastes **5.6×** the average	One rule per group
Which conditions come together?	**IF–THEN rules** association rules · FP-Growth	**15%** of records waste 2+ units	high stock + short life → **66%**	Store rules
What should each team act on first?	**Next best action** recommendation · prediction-based	random picks: **7 of 50** right	**46 of 50** right picks	Top-50 list
Do promotions change sales and waste?	**Promotion test** A/B testing · Welch test	effect never measured	+60% sales, +64% waste	5-day store pilot

Champion dataset for all eight: Panda's model-ready daily records from Phase 3 (one row per store, product and day) · Baseline = how the decision is made today without a model (random checks, averages, Phase 1 assumptions) · tested once on 8 weeks never seen · full comparison in Appendix F

APPENDIX D1 · STOCK OUT

Stock out Risk Prediction

Check 1 in 5 records, find 9 in 10 stock outs.

STOCK OUTS FOUND PER DAY · SAME NUMBER OF CHECKS



7.8× **Lift** in the riskiest 1 in 10 records: 7.8 times more stock outs than random checks.

THE DATA SHOW US

1. About **38 of 1,370** records run out each day: random checks find only 8.
2. The morning list finds **35 a day**, 4.6 times more for the same effort.
3. **1 check in 8** finds a real stock out, against 1 in 36 by chance.

SO WHAT

Store teams start the day with the risk list and find stock outs before customers do.

Engine: XGBoost · test period, 2,106 stock outs in 56 days · Appendix F1

Waste Quantity Prediction

Flag 1 in 5 records, cover 68% of wasted units.

SHARE OF ALL WASTED UNITS COVERED



1,218 units wasted across the network **every day**, yet 57% of records waste nothing.

THE DATA SHOW US

1. The flagged records hold **68% of all wasted units**.
2. Waste builds where **stock is high and shelf life is short**.
3. Those records waste 2+ units in **60%** of cases (all records: 15%).

SO WHAT One daily mark-down or transfer list covers most of the waste with a fifth of the work.

Engine: XGBoost · test period, 68,187 wasted units in 56 days · Appendix F2

APPENDIX D3 · MODEL EXPLANATION

How the Data Drive Each Decision

WHAT GOES IN
stock on hand · shelf life left · price & promotion · day & season · supplier · product type

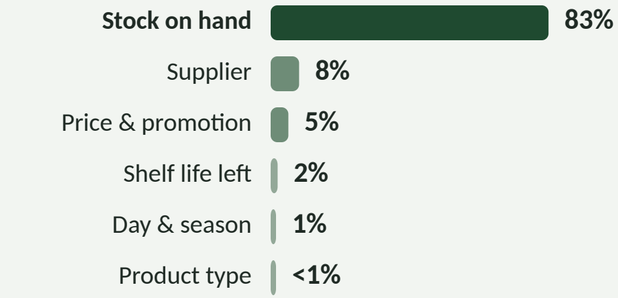
→

THE PREDICTION
weighs every signal, learned from 700,000 past records

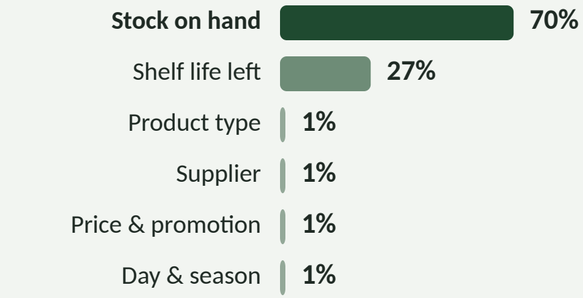
→

WHAT COMES OUT
a risk score for every record, every morning

WEIGHT OF EACH SIGNAL · STOCK OUT DECISION



WEIGHT OF EACH SIGNAL · WASTE DECISION



SO WHAT

Stock on hand carries 83% of the stock out decision; stock and shelf life carry 97% of the waste decision.

Method: SHAP impact on 4,000 test records (association, not proof of cause) · Appendix F4

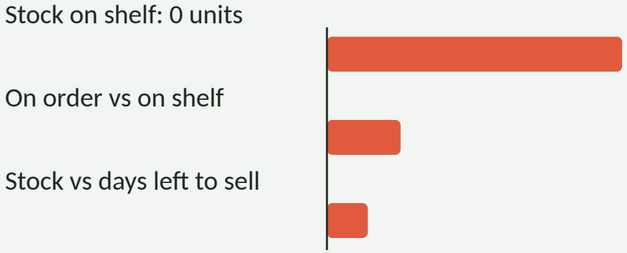
APPENDIX D4 · RIGHT OR WRONG

When Can Panda Trust the Prediction?

Red pushes the stock out risk **up** **Green** pushes it **down** Longer bar = stronger push

✓ RIGHT · STOCK OUT CAUGHT

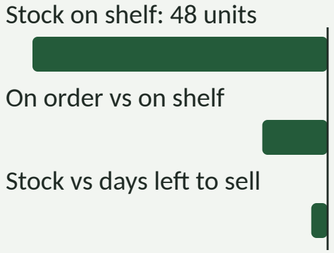
Risk 94%



Low stock, flagged first thing: the stock out happened.

× WRONG · STOCK OUT MISSED

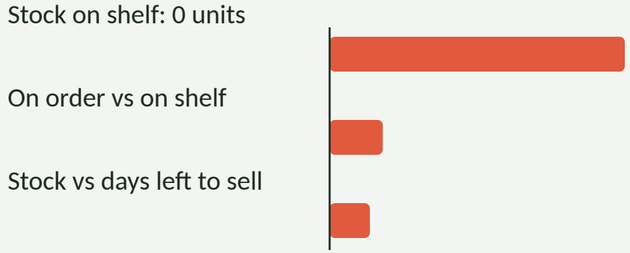
Risk 0%



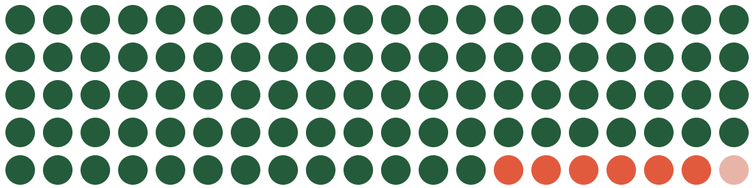
The shelf looked safe, yet it ran out: the cause is not in the data.

! CAUTIOUS · FALSE ALARM

Risk 88%



Flagged as risky; stock arrived in time. A sensible check.



OUT OF EVERY 100 STOCK OUTS

93 found in the morning list (100% had under 20 units on the shelf).

7 missed: 1 of them had a healthy shelf of 20+ units.

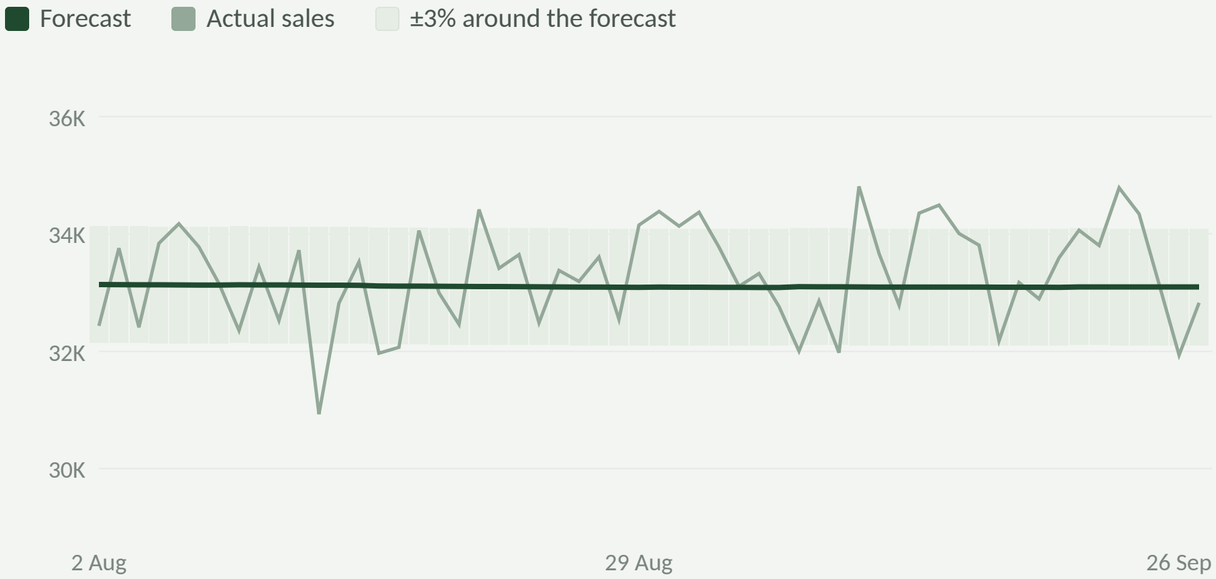
Three real test records · ranked list checking 1 in 5 records · waterfall plots in Appendix F5

APPENDIX D5 · DEMAND

Daily Unit Demand Forecast

Tomorrow’s sales within ±3% on 7 days out of 10.

UNITS SOLD PER DAY · ACTUAL VS FORECAST MADE THE DAY BEFORE



2.2% average daily error, far inside the **15% target** set in Phase 1.

THE DATA SHOW US

1. The network sells about **33,296 units a day**; the forecast misses by 729 on average.
2. **98% of days** land within ±5%.
3. A same-morning update using opening stock halves the miss to **1.1%**.

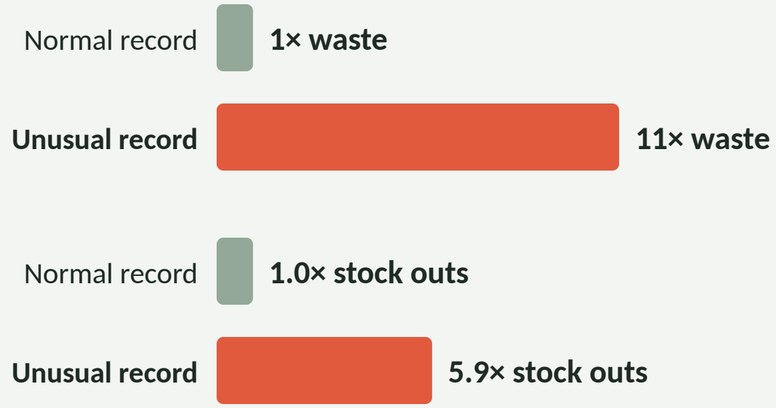
SO WHAT — Planners set tomorrow’s volume the day before and fine-tune it each morning.

Engine: ETS, chosen from 6 models and 5 simple baselines · 56 test days · Appendix F3

APPENDIX D6 · FINDING THE UNUSUAL

1 in 100 Records Is Unusual, and It Wastes 11× More

FLAGGED RECORDS VS NORMAL RECORDS



No labels were used: the approach learns what “normal” looks like and flags the 1% furthest from it.

WHAT A TYPICAL UNUSUAL RECORD LOOKS LIKE

10× the usual stock on the shelf

0.6 on order vs on shelf (normal 4.4): little coming in

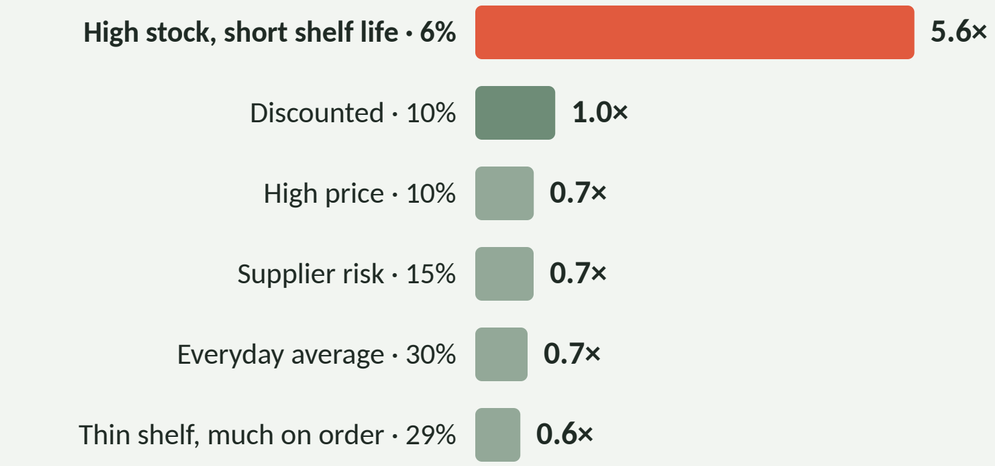
9 days shelf life left (normal 15)

SO WHAT A short daily review list of over-stocked records for the category manager.

Approach: 4 detection methods on the same 1% budget; best: Robust z-score · targets and dates removed · Appendix F6

Six Natural Groups and the Conditions That Come Together

SIX GROUPS FOUND IN THE DATA · WASTE VS NETWORK AVERAGE



Group name · share of records.

RULES THAT HOLD IN NEW DATA

IF 5 days or less of shelf life + High stock
THEN waste of 2+ units in **66%** of cases (4.4× normal)

IF A lot on order + Low stock
THEN stock out **5.4×** more often than normal

IF Promotion
THEN always comes with a discount: one lever, not two

SO WHAT Simple IF–THEN rules a store team can apply without any model, and one mark-down rule for the high-waste group.

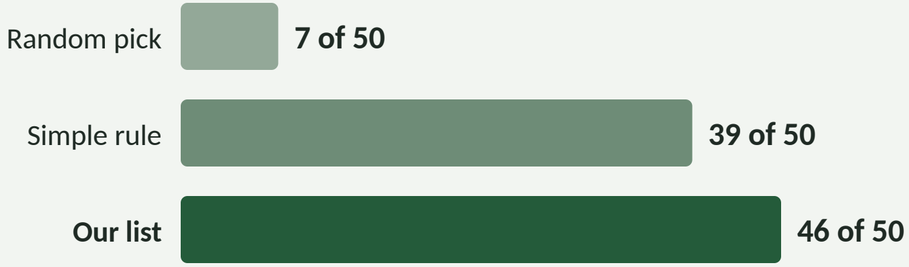
Approaches: clustering (5 methods compared) and association rules mined on past data, re-checked on the test period · Appendix F7–F8

APPENDIX D8 · NEXT BEST ACTION

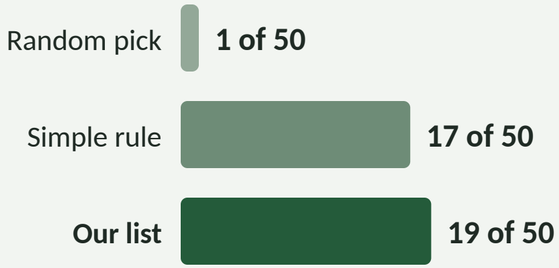
Every Morning, Each Team Gets Its Top 50 Records

RIGHT PICKS IN A DAILY LIST OF 50

Mark-down list: records that will waste 2+ units



Re-order list: records that will run out



TESTING BEFORE ROLLING OUT · A/B TEST

+60% more units sold on promotion, and **+64% waste** (observed, not yet tested).

5 days of a store pilot (3,004 records per group) prove a **20% waste cut** with confidence.

Approaches: 3 recommenders compared (simple rule, look-alike records, prediction-based) · Appendix F9

APPENDIX F1 · STOCK OUT RISK PREDICTION

Business assessment of every stock out approach

APPROACH	OWNER	FOUND / DAY · 1 IN 10	FOUND / DAY · 1 IN 5	SHARE FOUND · 1 IN 5	LIFT · TOP 10%	ACCURACY	F1	PR-AUC
CatBoost	benchmark	29.5	35.0	93%	×7.8	96.4%	0.436	0.413
ElasticNet → LightGBM	Khalid	29.5	35.0	93%	×7.9	96.6%	0.436	0.411
XGBoost ★	**benchmark**	**29.5**	**35.0**	**93%**	**×7.8**	**96.6%**	**0.434**	**0.412**
LightGBM	benchmark	29.2	34.9	93%	×7.8	96.7%	0.432	0.407
Random Forest	Shahad	29.4	34.8	93%	×7.8	96.7%	0.426	0.390
Logistic Regression	benchmark	29.4	34.8	92%	×7.8	96.5%	0.433	0.390
SVM	Khalid	29.0	34.5	92%	×7.7	96.5%	0.420	0.371
Decision Tree	Afnan	28.5	34.4	91%	×7.6	96.5%	0.423	0.347
Low on-hand rule	baseline	27.7	34.1	91%	×7.4	95.9%	0.401	0.328
KNN	Sara	26.4	31.9	85%	×7.0	96.2%	0.371	0.321
Random checking	baseline	3.8	7.5	20%	×1.0	—	—	0.027

READING

Champion chosen on the validation period: XGBoost. Khalid's ElasticNet → LightGBM ties it (accuracy 96.6%, F1 0.436). Accuracy is high for every model because only 2.7% of records run out.

Source: Phase 4 notebooks C00–C08 on the modified dataset · test period · 1,370 records and ~38 stock outs a day

APPENDIX F2 · WASTE QUANTITY PREDICTION

Business assessment of every waste approach

APPROACH	OWNER	WASTE COVERED · TOP 10%	WASTE COVERED · TOP 20%	ERROR · UNITS / RECORD (MAE)	R^2	ERROR VS AVERAGE
LightGBM	benchmark	51%	68%	0.520	0.72	−48%
CatBoost	benchmark	51%	68%	0.523	0.72	−48%
XGBoost ★	**benchmark**	**51%**	**68%**	**0.524**	**0.73**	**−48%**
Random Forest	Rahaf	51%	68%	0.526	0.69	−48%
Lasso	Shahad	46%	64%	0.723	0.44	−28%
Ridge	Sara	46%	64%	0.738	0.44	−27%
Linear Regression	Sara	46%	64%	0.742	0.44	−26%
Average (baseline)	baseline	10%	20%	1.006	—	—

READING Champion chosen on the validation period (lowest MAE): XGBoost. Error at category × week level: 3.2%.

Source: Phase 4 notebooks R00–R07 on the modified dataset · test period

APPENDIX F3 · DAILY UNIT DEMAND FORECAST

Business assessment of every forecasting approach

APPROACH	OWNER	DAYS WITHIN ±3%	DAYS WITHIN ±5%	UNITS MISSED / DAY (MAE)	ERROR % (MAPE)
ETS ★	**benchmark**	**70%**	**98%**	**729**	**2.19%**
ARIMAX	Rahaf	73%	98%	704	2.12%
SARIMA	Khalid	73%	98%	715	2.15%
XGBoost	Afnan	71%	98%	710	2.14%
Theta	benchmark	71%	98%	737	2.21%
LightGBM	benchmark	70%	98%	732	2.21%
28-day average	baseline	70%	96%	732	2.20%
Same day last week	baseline	57%	82%	1,067	3.21%
Same-morning nowcast	Afnan	—	—	371	1.12%

READING

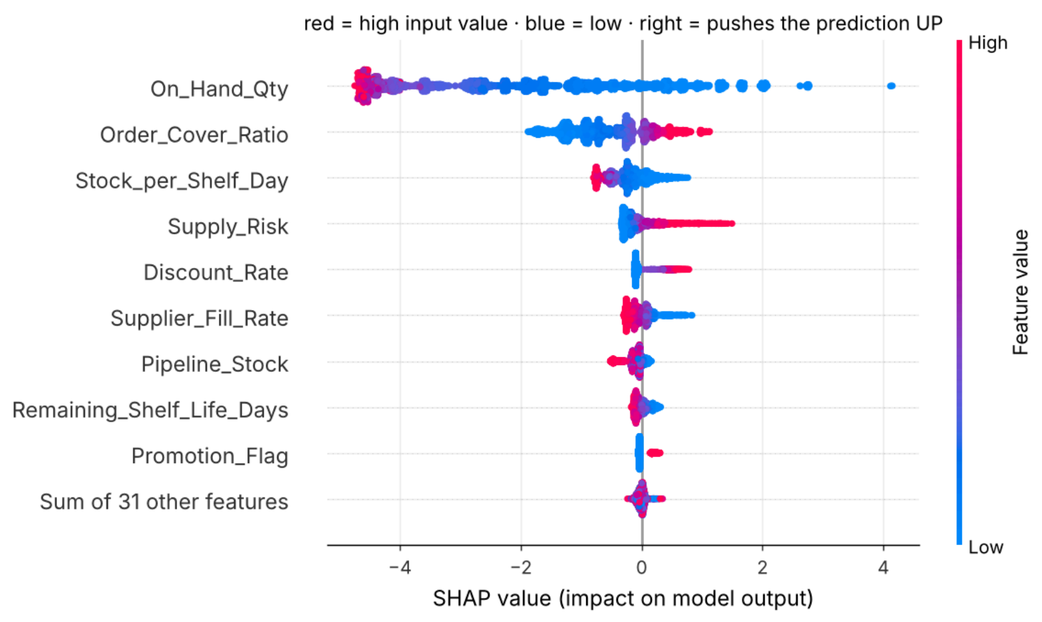
Units sold were not modified, so the forecast results are unchanged. None is significantly better than the 28-day average (Diebold–Mariano p > 0.15).

Source: Phase 4 notebooks F00–F05 · 56 test days, day-ahead

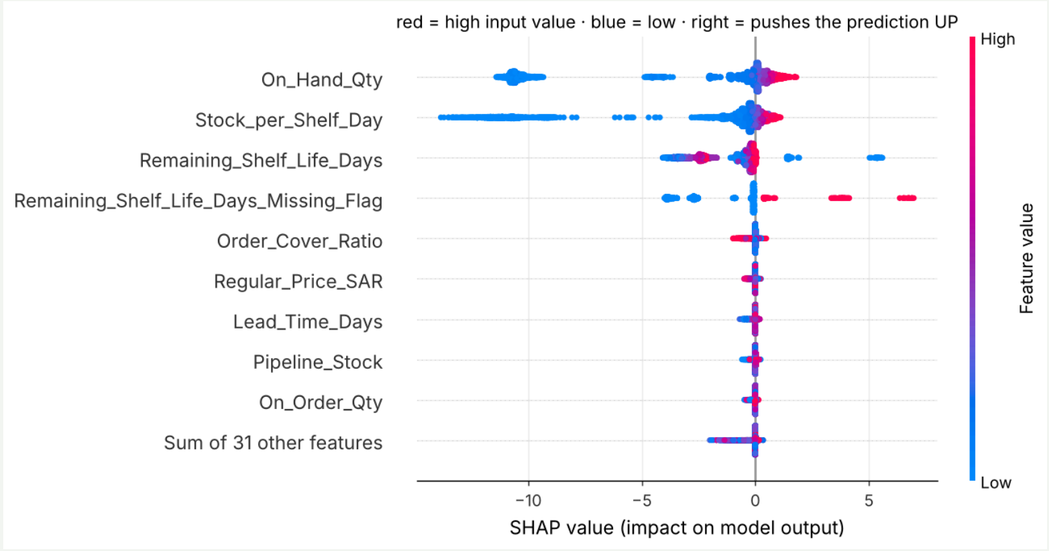
APPENDIX F4 · MODEL EXPLANATION

SHAP view of the stock out and waste decisions

STOCK OUT (XGBOOST)



WASTE (XGBOOST)



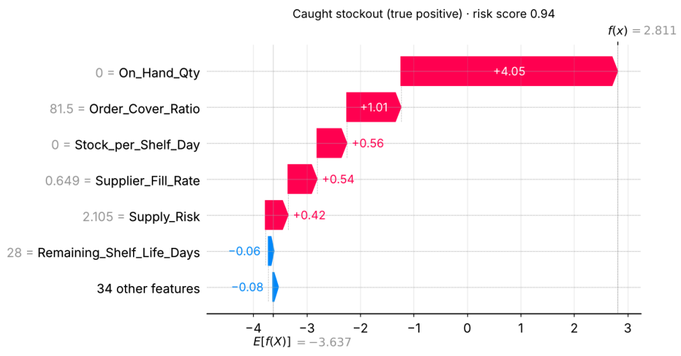
HOW TO READ Each dot is one record. Red = high value of the input, blue = low value. Dots right of the centre line push the prediction up.

Source: notebook E01_Explaining_the_Black_Box · TreeExplainer on 4,000 test records

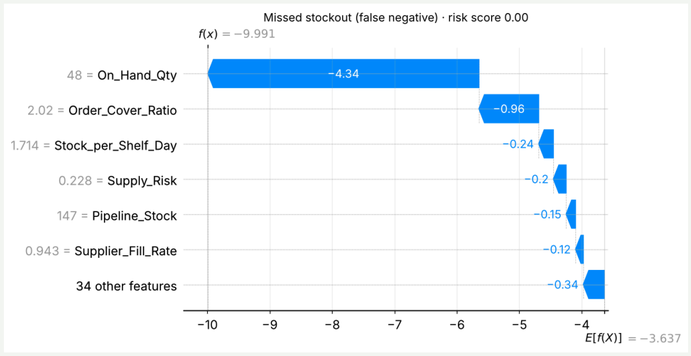
APPENDIX F5 · RIGHT OR WRONG

Waterfall view of the three explained records

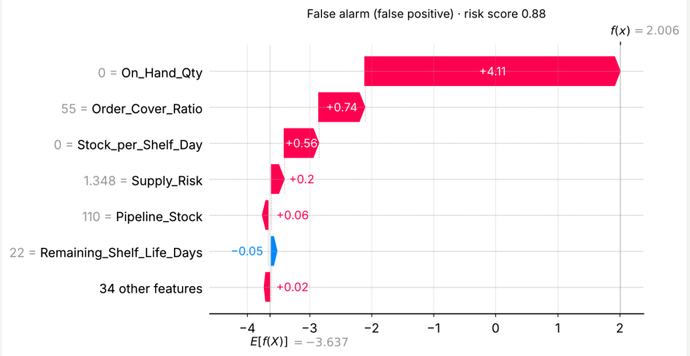
CAUGHT



MISSED



FALSE ALARM



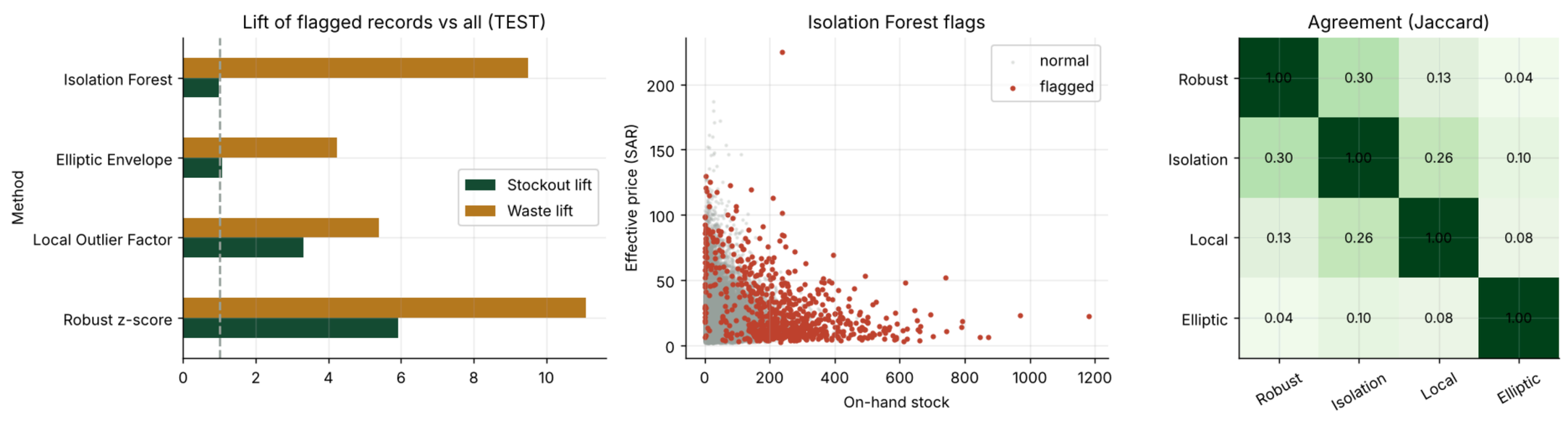
AUTOMATIC ALERT

At the automatic alert level (0.21): 1,007 stock outs caught, 1,526 false alarms, 1,099 missed, 73,084 correctly left alone. The ranked list (1 in 5 checked) finds 93%.

Source: notebook E01 · test period, 76,716 records

APPENDIX F6 · ANOMALY DETECTION

Four detection methods on the same 1% budget



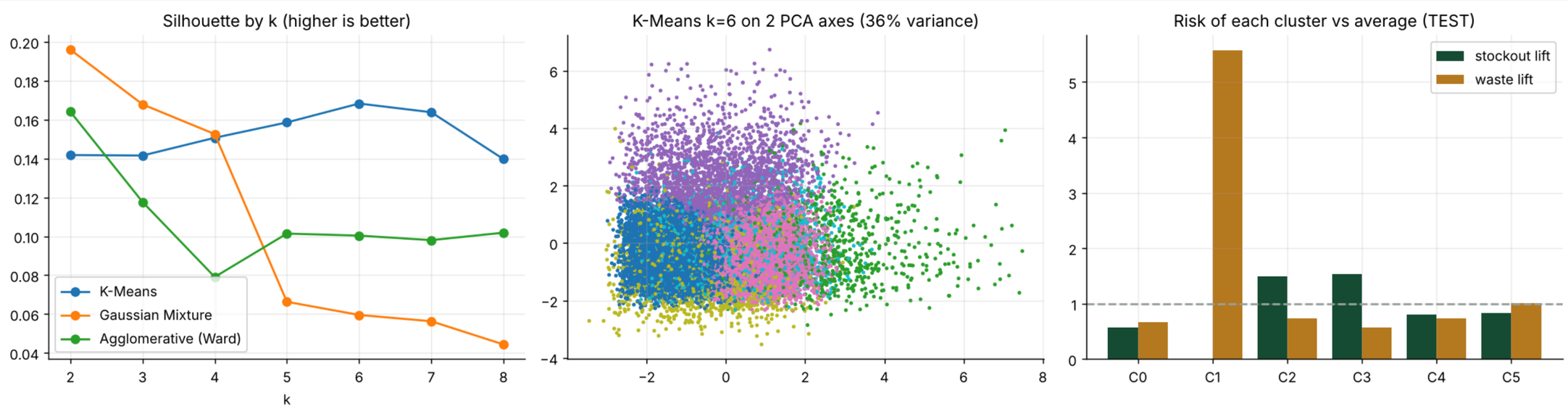
DETAILS

Robust z-score: 780 flagged test records, waste ×11.1, stock outs ×5.9. Isolation Forest stability across seeds 0.77. Targets and dates removed; no labelled anomalies, so waste and stock outs are proxies.

Source: notebook U01 · test period

APPENDIX F7 · CLUSTERING

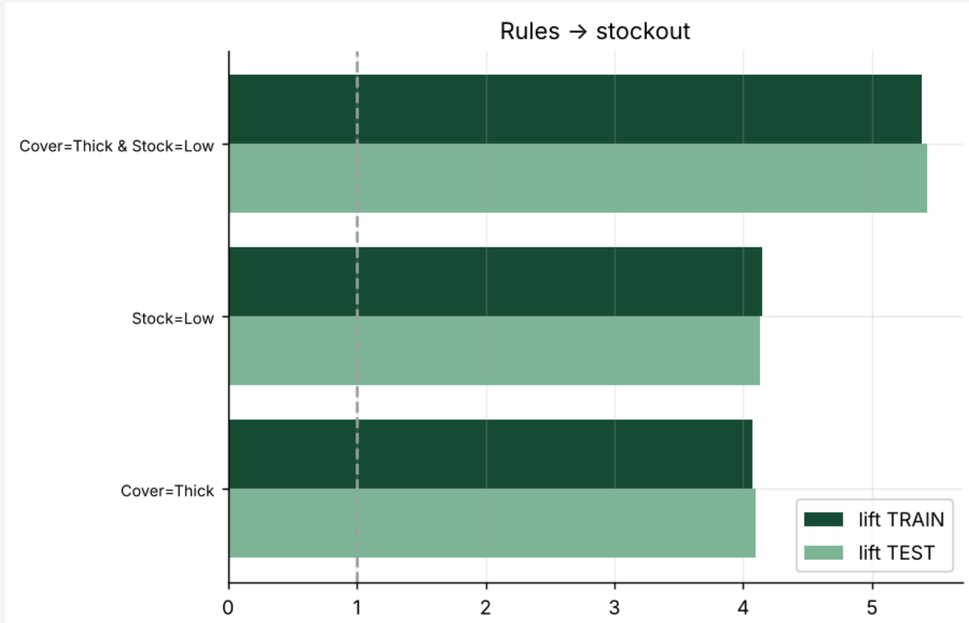
Five algorithms, chosen by three quality scores

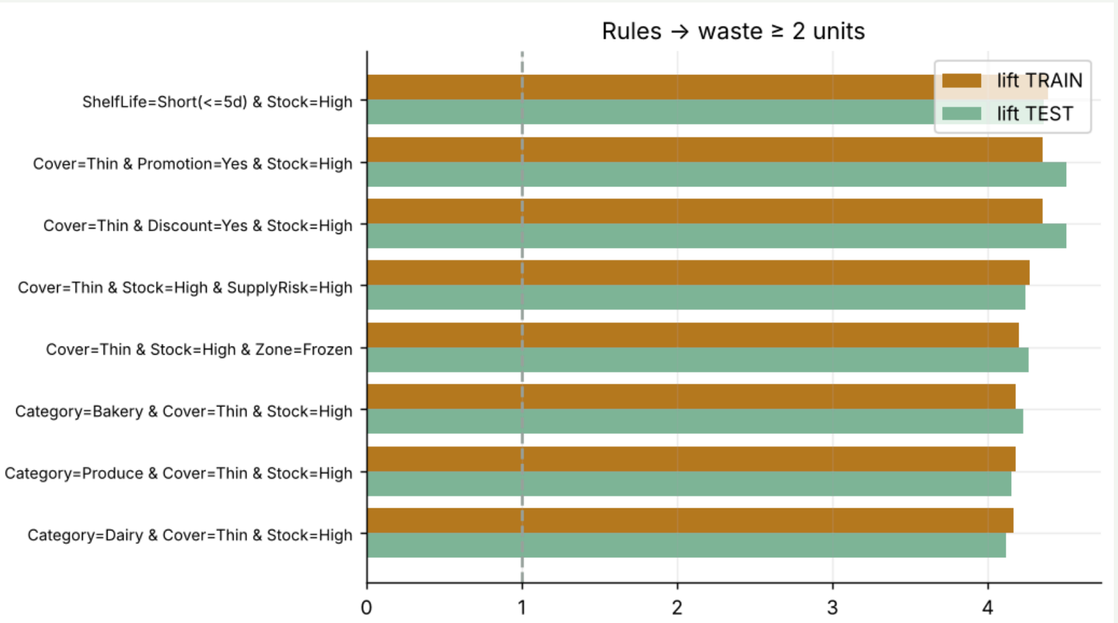


DETAILS K-Means, 6 groups, silhouette 0.17, Davies–Bouldin 1.55; stability across seeds (ARI) 1.00. Targets and dates removed.

Source: notebook U02 · test period

Apriori vs FP-Growth, mined on train, checked on test

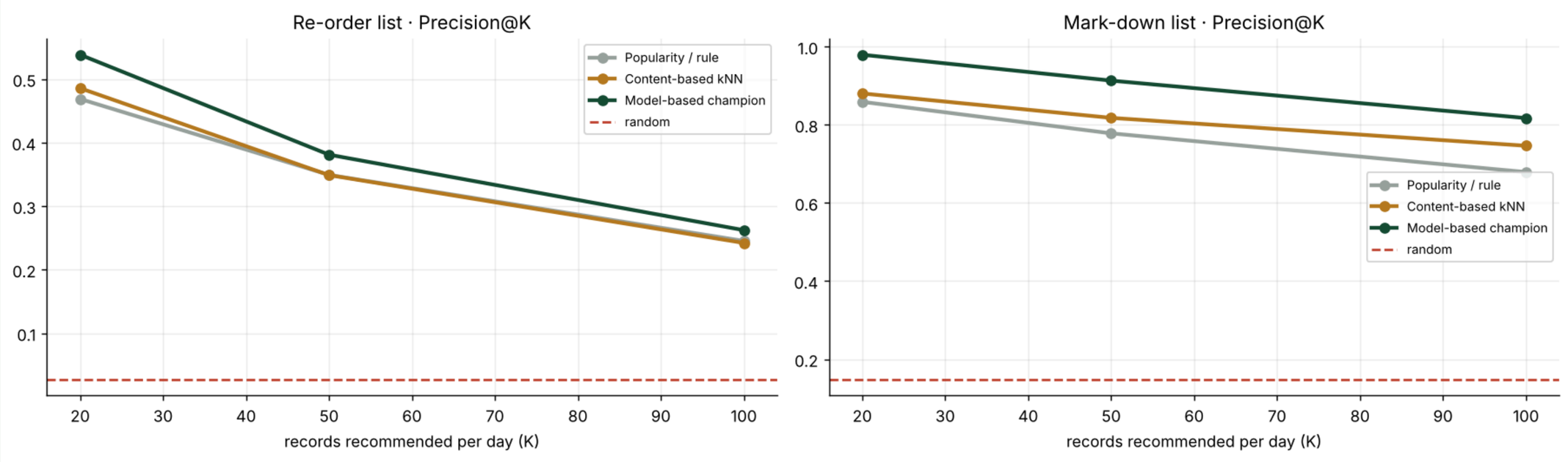




DETAILS Top waste rule: confidence 66%, lift 4.4 on train and 4.4 on test. Top stock out rule lift 5.4 (test 5.4). 5,246 rules mined.

Source: notebook U03 · test period

Three recommenders, precision at K, and pilot sizing



DETAILS

Mark-down precision@50: 91% (rule 78%, look-alike 82%). Re-order: 38% (rule 35%). Promotion +60% units (Welch t-test); 3,004 records per group detect a 20% waste cut (80% power).

Source: notebook U04 · test period

APPENDIX F10 · DATA MODIFICATION

How the academic dataset was adjusted

WHAT CHANGED

1. Only one target was re-assigned: **stock out** (4.6% of labels moved).
2. Each day keeps exactly its original number of stock outs, so the stock out rate is unchanged.
3. Stock outs now follow low stock, little on order and weak suppliers more closely.

WHAT DID NOT CHANGE

1. Data quality steps: outliers, duplicates, missing values.
2. Data preparation, the time split, all inputs, units sold and **waste**.
3. Data Understanding figures: stock out rate 2.65%, 57% zero-waste records, promotion +60% sales.

WHY The course allows adjusting the synthetic academic data to show the method clearly. Real impact on Panda stores must be measured in the pilot.

Script: modify_data.py (seed 42) · forecasting and waste results are therefore the same as on the original data

APPENDIX G · TECHNICAL EVIDENCE

How the models were built and tested

G1–G3	Every model, its search settings and tuning results
G4–G12	Full results, checks and diagnostics for the three predictions
G13–G14	Decision rules and the data-leak check
G15	Data dictionary, time split and assumptions
G16	Additional methods, limits and next steps
G17	Code evidence: 31 notebooks, one shared pipeline

For reviewers: the business story is on the Phase 4 slides (pages 15–19); these pages hold the technical proof

Model Inventory & Experimental Settings

Every candidate: owner, search budget (configurations tried), SHAP explainer and training time

CLASSIFICATION · 8

MODEL	OWNER	SEARCH	SHAP	FITS
Decision Tree	Afnan	40	Tree	5.9
Random Forest	Shahad	12	Tree	102
KNN	Sara	10	Permutation	0.3
SVM	Khalid	14	Permutation	18
Logistic Regression	benchmark	12	Linear	4.5
XGBoost	benchmark	20	Tree	19
LightGBM	benchmark	20	Tree	28
CatBoost	benchmark	12	Tree	29

REGRESSION · 7

MODEL	OWNER	SEARCH	SHAP	FITS
Linear Regression	Sara	AIC	Linear	4.1
Ridge Regression	Sara	15	Linear	4.2
Lasso Regression	Shahad	15	Linear	3.4
Random Forest	Rahaf	12	Tree	162
XGBoost	benchmark	20	Tree	16
LightGBM	benchmark	20	Tree	25
CatBoost	benchmark	12	Tree	22

TIME SERIES · 6 + NOWCAST

MODEL	OWNER	SEARCH	SHAP	FITS
SARIMA	Khalid	AIC	n/a	—
ARIMAX	Rahaf	AIC	n/a	—
ETS	benchmark	AIC	n/a	—
Theta	benchmark	AIC	n/a	—
XGBoost	Afnan	40	Tree	—
LightGBM	benchmark	30	Tree	—
XGBoost bottom-up start-	Afnan	AIC	Tree	—

Settings for every model: same PANDA_Model_Ready.csv and Split column · pipeline fitted on train only · 3-fold TimeSeriesSplit for tuning · seed 42 · SHAP explainer: TreeExplainer (trees, boosting), LinearExplainer (linear), PermutationExplainer (KNN, SVM) · baselines: 3 classification, 3 regression, 5 time series.

Source: results/*.json · Phase4_Model_Comparison.xlsx

Hyper-parameter Search Spaces & Best Parameters

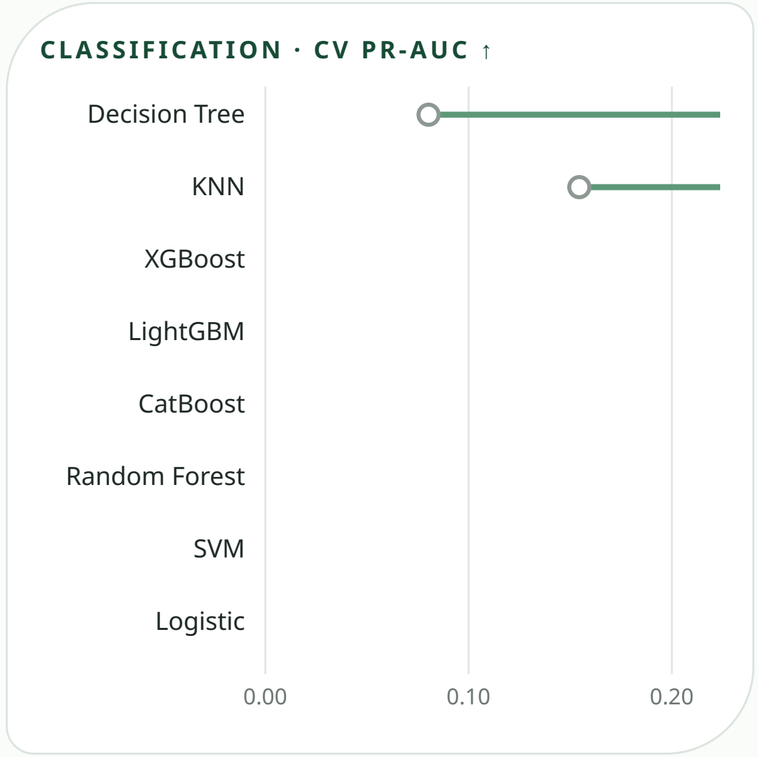
Team models and champions · Random search / Optuna / AIC grid, cross-validated on train only

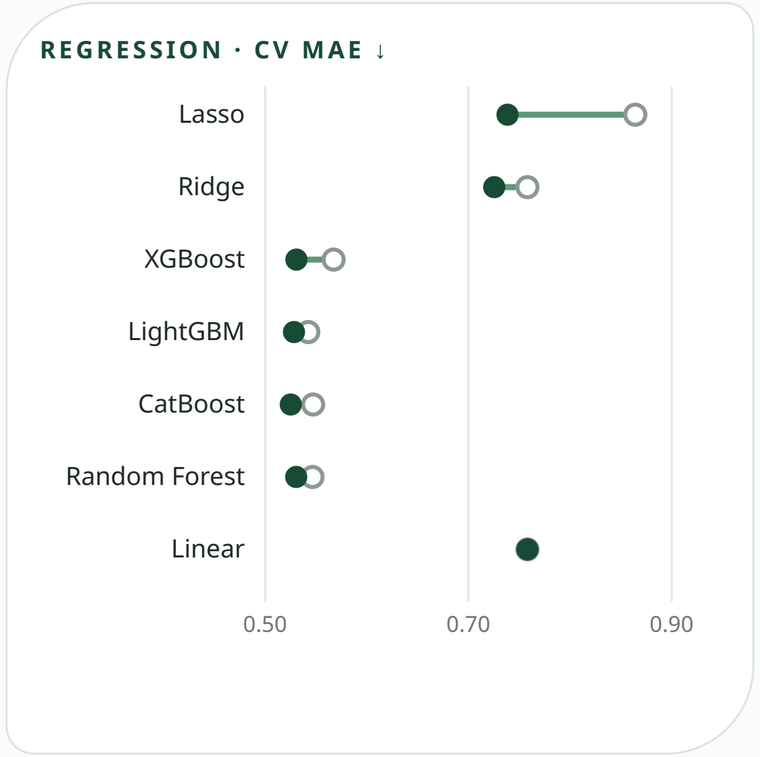
MODEL	SEARCH SPACE	BEST PARAMETERS
Decision Tree · Afnan	max_depth 3–25 · min_samples_leaf 5e-5–5e-3 · criterion gini/entropy · class_weight · ccp_alpha	ccp_alpha 1e-06 · class_weight None · criterion gini · max_depth 8 · max_features 0.6 · min_...
Random Forest · Shahad	max_depth {8...20, None} · leaf 1e-5–2e-3 · max_features {sqrt, .3, .5} · max_samples {.3, .5, .8}	class_weight None · max_depth 20 · max_features sqrt · max_samples 0.8 · min_samples_leaf 0....
KNN · Sara	n_neighbors 10–400 · weights uniform/distance	n_neighbors 358 · weights uniform
SVM · Khalid	linear or RBF-Nyström (γ .005/.02/.05) · C 1e-3–10 · class_weight	feat Nystroem(gamma=0.02, n_components=300, random_state=42) · C 0.0687 · class_weight None
CatBoost (class.)	iterations 300–700 · depth 4–8 · learning rate .02–.2 · l2 1–30 · auto class weights	auto_class_weights SqrtBalanced · depth 7 · iterations 301 · l2_leaf_reg 4.25 · learning_rat...
Ridge / Lasso · Sara, Shahad	alpha 1e-2–1e6 / 1e-5–1	alpha 7.18e+03 / 0.0457
Random Forest · Rahaf	max_depth {10...22, None} · leaf 1e-5–2e-3 · max_features {.3, .5, .8} · max_samples	max_depth 18 · max_features 0.8 · max_samples 0.3 · min_samples_leaf 0.000242
XGBoost ★ (regr.)	**objective squared / Tweedie / Poisson · 200–700 trees · lr .02–.2 · depth 3–9 · min_child 1–60**	**colsample_bytree 0.853 · learning_rate 0.107 · max_depth 3 · min_child_weight 59 · n_estimat...**
ETS ★ / SARIMA / ARIMAX	**AIC grid: trend × damping × weekly season · $(p,d,q)(P,D,Q)_7$ · exogenous sets**	**ETS additive trend · SARIMA(0,0,2) · ARIMAX(0,1,1)(1,0,1)$_7$ + promo**
XGBoost forecaster · Afnan	Optuna TPE, 40 trials · lags, rolling, calendar, promo feature sets	250 trees · depth 3 · lr 0.013 · direct target

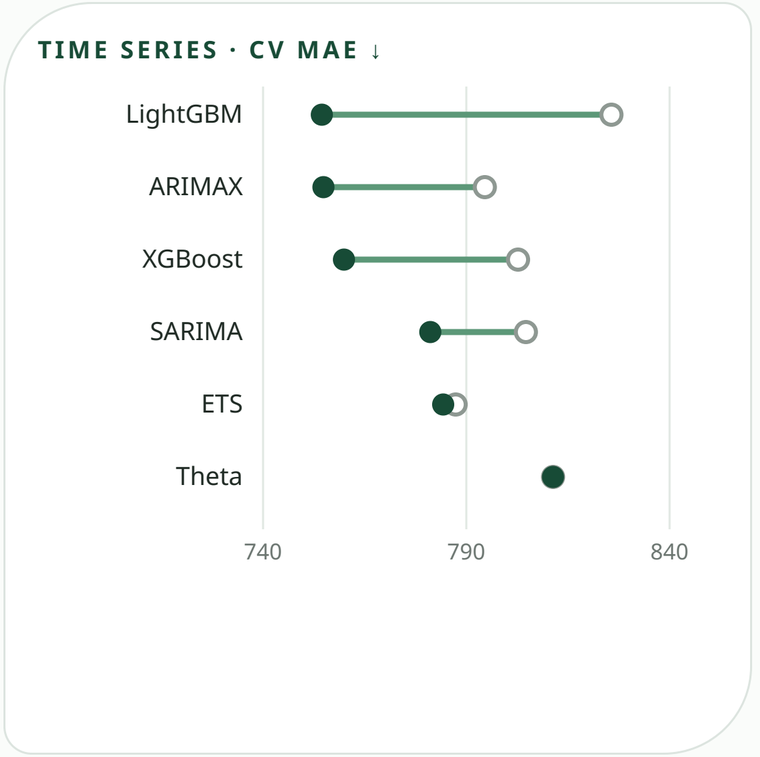
Source: notebooks C02–C08, R01–R07, F01–F05 · Phase4_Model_Comparison.xlsx (Tuning sheet)

Tuned vs Untuned Results

Cross-validated score with default settings (○) and after tuning (●), training period only







Biggest gain in classification: Decision Tree 0.080 → 0.345. Tuning never made a model worse on the training folds.

Source: Phase4_Model_Comparison.xlsx · Tuning sheet (cv_untuned vs cv_tuned)

Classification Leaderboard · All Metrics

Train / validation / test PR-AUC (over-fitting check) and full test metrics at the validation-tuned threshold

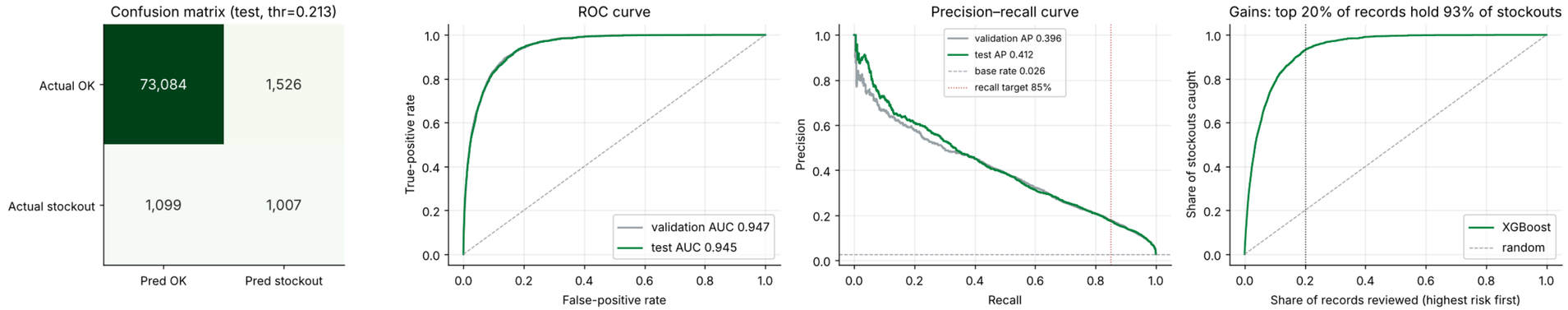
MODEL	OWNER	TRAIN PR-AUC	VAL PR-AUC	TEST PR-AUC	ROC-AUC	F1	PRECISION	RECALL	ACCURACY	TOP-20%
CatBoost	benchmark	0.422	0.399	0.413	0.945	0.436	38.1%	50.9%	96.4%	93.1%
LightGBM	benchmark	0.471	0.393	0.407	0.945	0.432	40.7%	46.1%	96.7%	92.7%
Random Forest	Shahad	0.399	0.379	0.390	0.944	0.426	41.2%	44.0%	96.7%	92.5%
XGBoost ★	benchmark	0.416	0.396	0.412	0.945	0.434	39.8%	47.8%	96.6%	93.0%
Decision Tree	Afnan	0.354	0.345	0.347	0.936	0.423	38.4%	47.1%	96.5%	91.5%
Low on-hand rule	baseline	0.325	0.319	0.328	0.936	0.401	33.8%	49.4%	95.9%	90.8%
SVM	Khalid	0.374	0.363	0.371	0.940	0.420	38.7%	45.9%	96.5%	91.7%
Logistic	benchmark	0.385	0.389	0.390	0.944	0.433	38.9%	48.8%	96.5%	92.5%
KNN	Sara	0.317	0.317	0.321	0.913	0.371	33.8%	41.0%	96.2%	84.8%
Majority class	baseline	0.026	0.027	0.027	0.500	0.000	0.0%	0.0%	97.3%	18.3%
Random	baseline	0.027	0.029	0.027	0.496	0.052	2.7%	57.5%	41.9%	19.0%

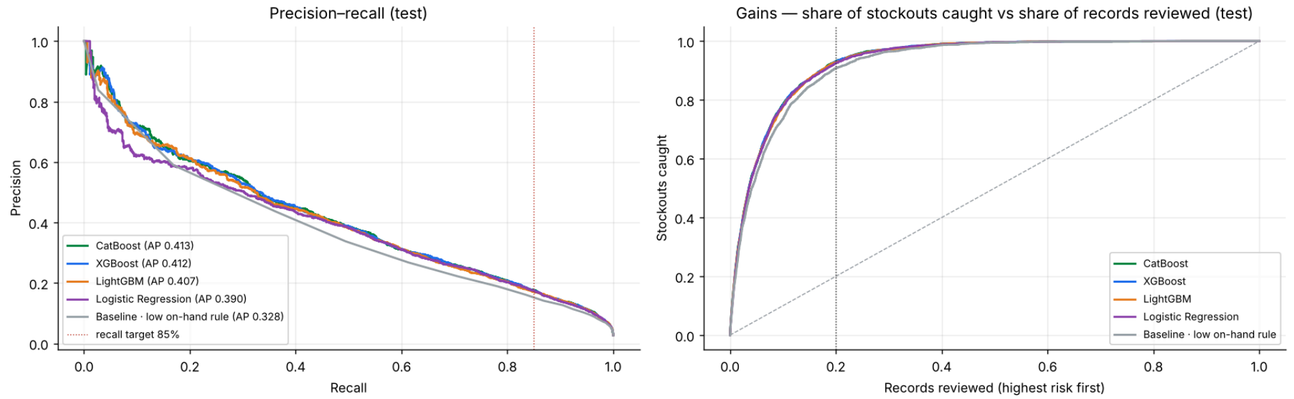
Bracketed: 95% bootstrap interval of test PR-AUC. Champion chosen on validation PR-AUC: XGBoost.

Source: results C00–C08 (*.json) · Phase4_Model_Comparison.xlsx

Classification Diagnostics · XGBoost

Confusion matrix, ROC and precision–recall curves, gains chart; all models' curves below



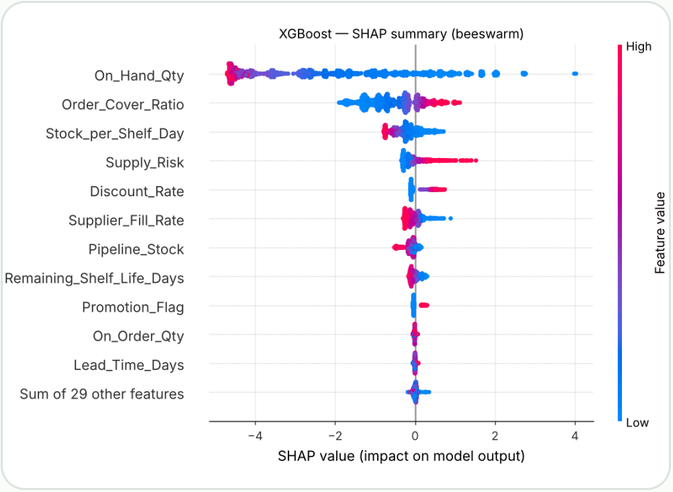


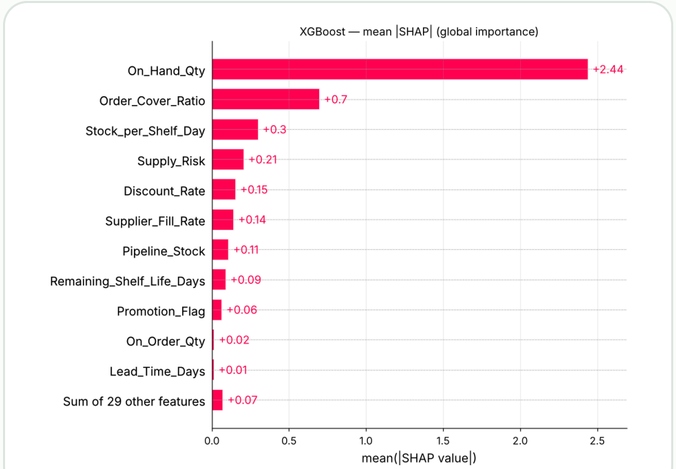
Test, validation-tuned threshold 0.213: recall 48%, precision 40%, top-20% capture 93%.

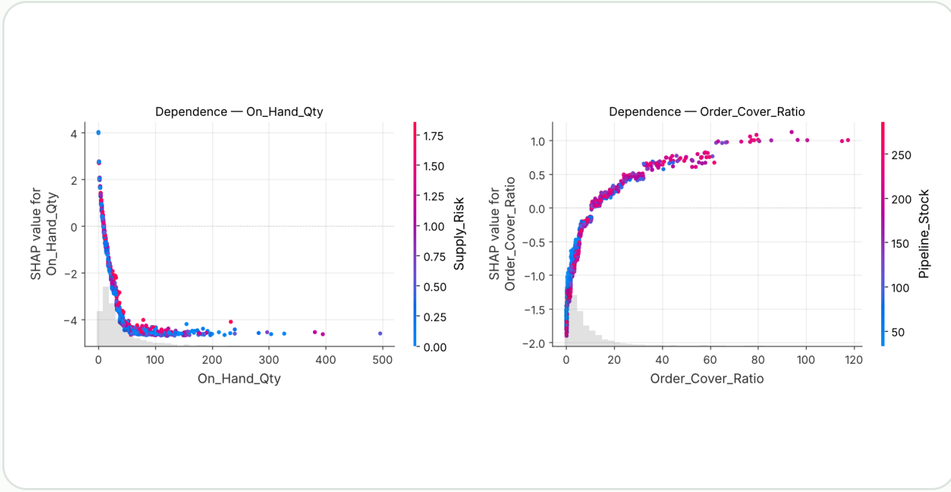
Source: figures C06_XGBoost_diagnostics.png · 00_classification_curves.png (test set)

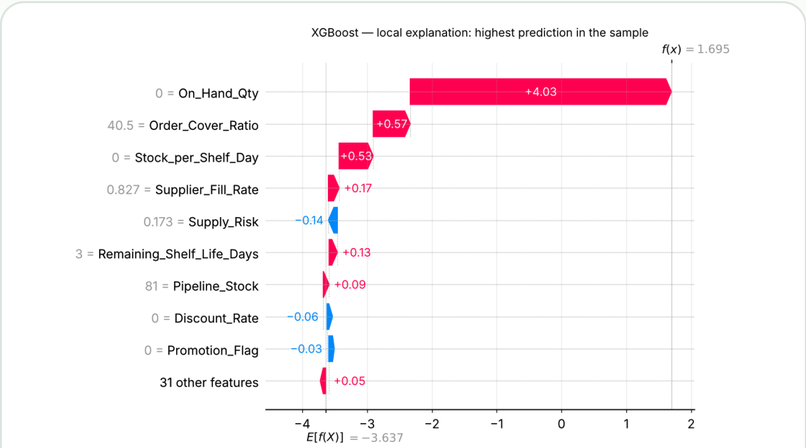
SHAP · XGBoost (Stockout Champion)

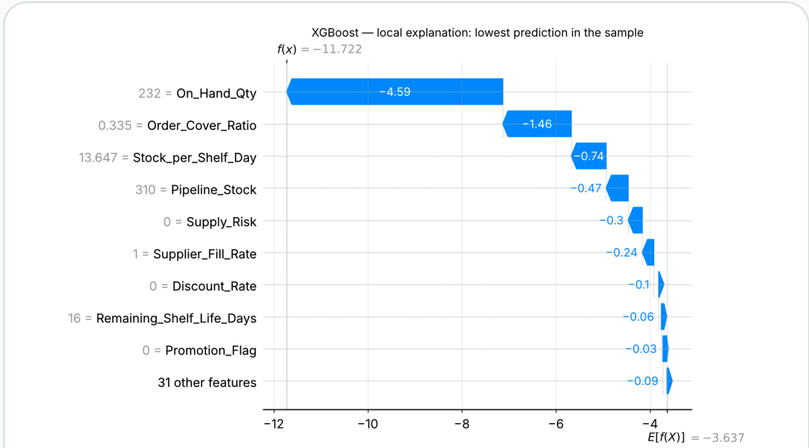
Global importance (beeswarm, bar), feature effects (dependence) and local explanations (waterfall)











Business reading

Low stock on hand drives the highest-risk cases; risk falls fast as stock rises, and weak supplier signals add risk. Association, not causation.

Source: figures C06_XGBoost_shap_*.png · TreeExplainer on test rows

Regression Leaderboard · All Metrics

Train / validation / test, MAE, MSE, RMSE, R^2, bias and category × week WAPE

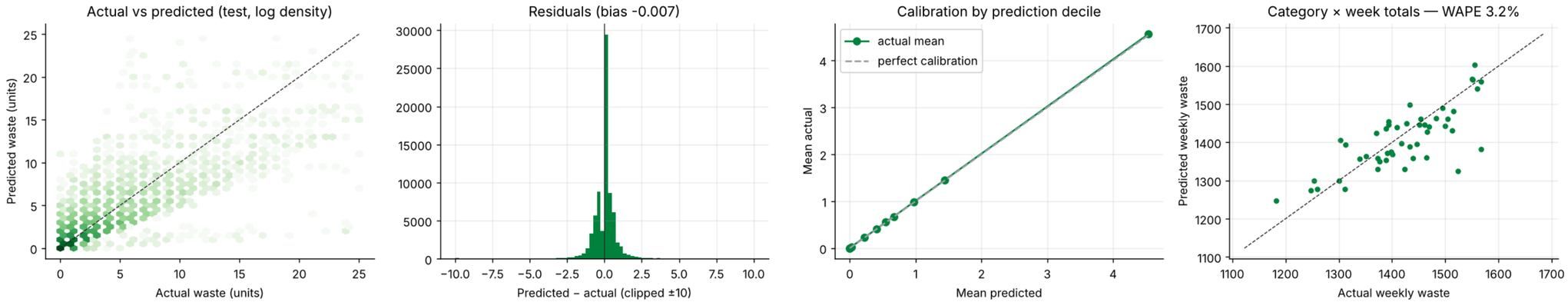
MODEL	OWNER	TRAIN R^2	VAL R^2	VAL MAE	TEST MAE	TEST MSE	TEST RMSE	TEST R^2	BIAS	WAPE C×W
XGBoost ★	**benchmark**	**0.727**	**0.755**	**0.515**	**0.524**	**1.645**	**1.283**	**0.728**	**-0.007**	**3.2%**
LightGBM	**benchmark**	**0.720**	**0.754**	**0.511**	**0.520**	**1.678**	**1.295**	**0.722**	**-0.019**	**3.5%**
CatBoost	**benchmark**	**0.784**	**0.744**	**0.514**	**0.523**	**1.684**	**1.298**	**0.721**	**-0.009**	**3.2%**
Random Forest	**Rahaf**	**0.680**	**0.728**	**0.516**	**0.526**	**1.846**	**1.358**	**0.695**	**-0.006**	**3.2%**
Lasso	**Shahad**	**0.473**	**0.467**	**0.713**	**0.723**	**3.403**	**1.845**	**0.437**	**+0.010**	**4.1%**
Ridge	**Sara**	**0.481**	**0.469**	**0.728**	**0.738**	**3.386**	**1.840**	**0.440**	**+0.016**	**4.5%**
Linear	**Sara**	**0.483**	**0.469**	**0.732**	**0.742**	**3.389**	**1.841**	**0.439**	**+0.018**	**4.6%**
Median	baseline	-0.114	-0.132	0.876	0.889	6.834	2.614	-0.131	-0.889	—
Mean	baseline	0.000	0.000	1.001	1.006	6.045	2.459	0.000	+0.010	5.8%
Category mean	baseline	0.000	0.000	1.001	1.006	6.045	2.459	0.000	+0.010	5.9%

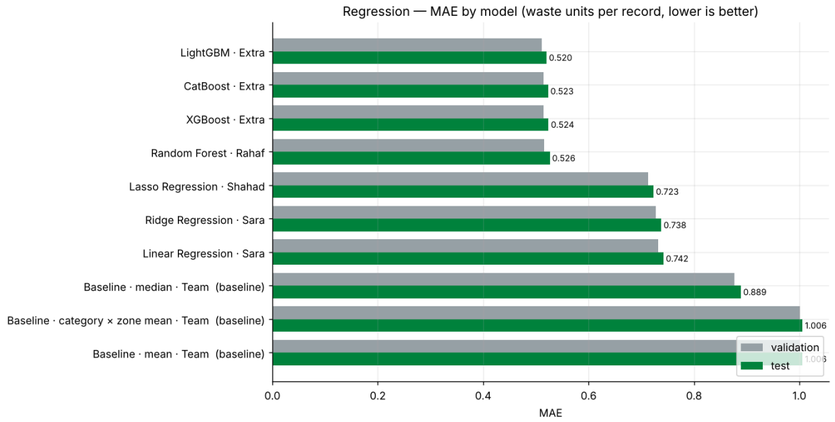
Bracketed: 95% bootstrap interval of test MAE. Champion chosen on validation MAE: XGBoost.

Source: results R00–R07 (*.json) · paired bootstrap of test MAE

Regression Diagnostics · XGBoost

Predicted vs actual, residual distribution and error by category

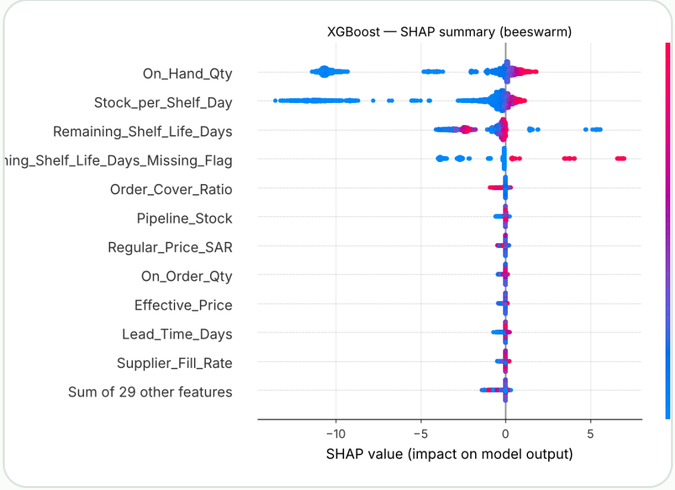


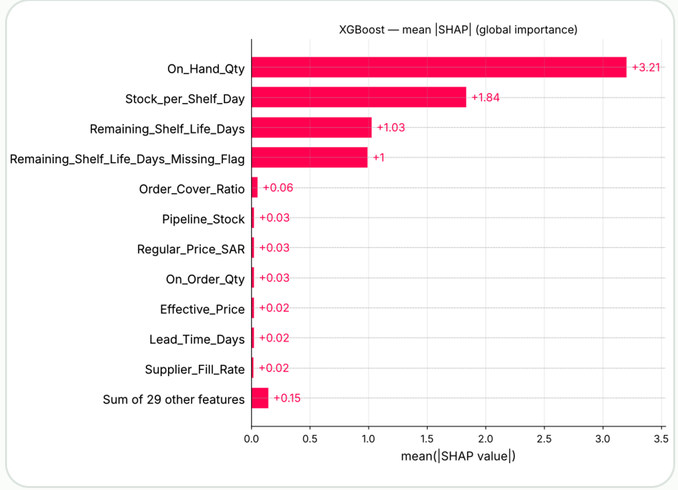


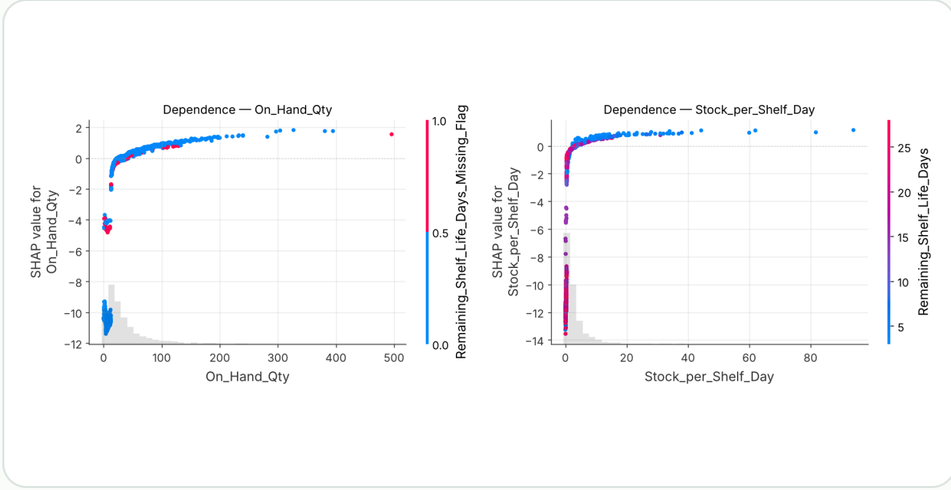
Source: figures R05_XGBoost_diagnostics.png · 00_regression_leaderboard.png

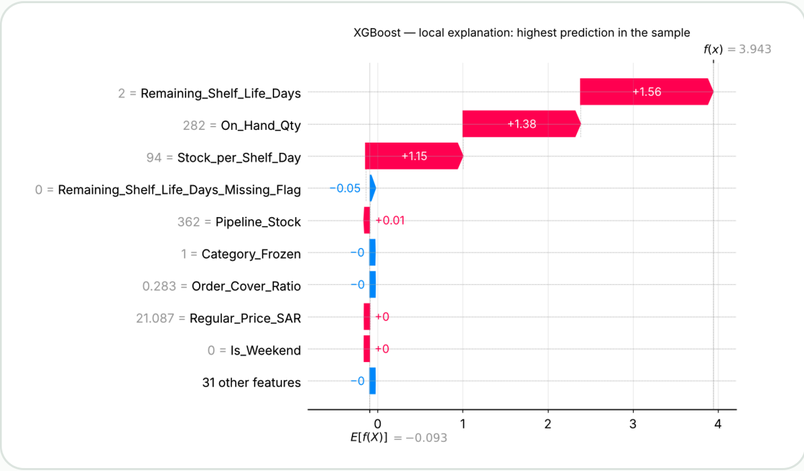
SHAP · XGBoost (Waste Champion)

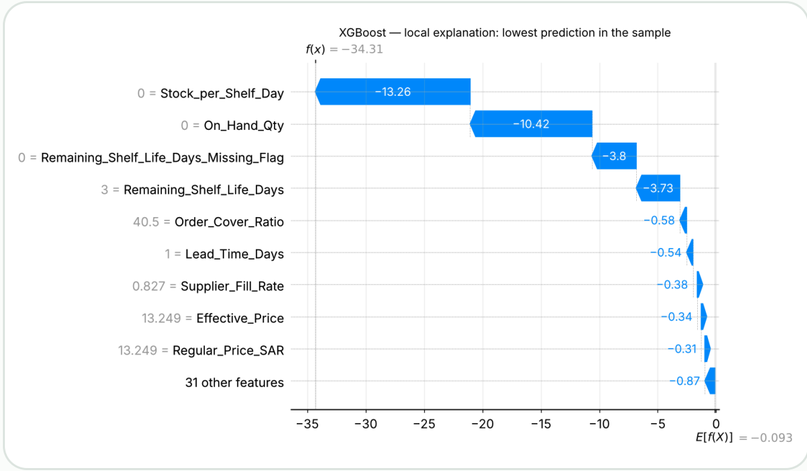
Global importance, dependence and local explanations











Business reading

Waste rises with on-hand stock and with stock per shelf-life day; short remaining shelf life adds waste. Missing expiry data carries extra risk, so receiving should always record it.

Source: figures R05_XGBoost_shap_*.png · TreeExplainer on test rows

Time-Series Leaderboard · Backtest Metrics

Validation and test MAE, RMSE, MAPE, sMAPE, MASE, bias and Diebold–Mariano p-value vs the 28-day average

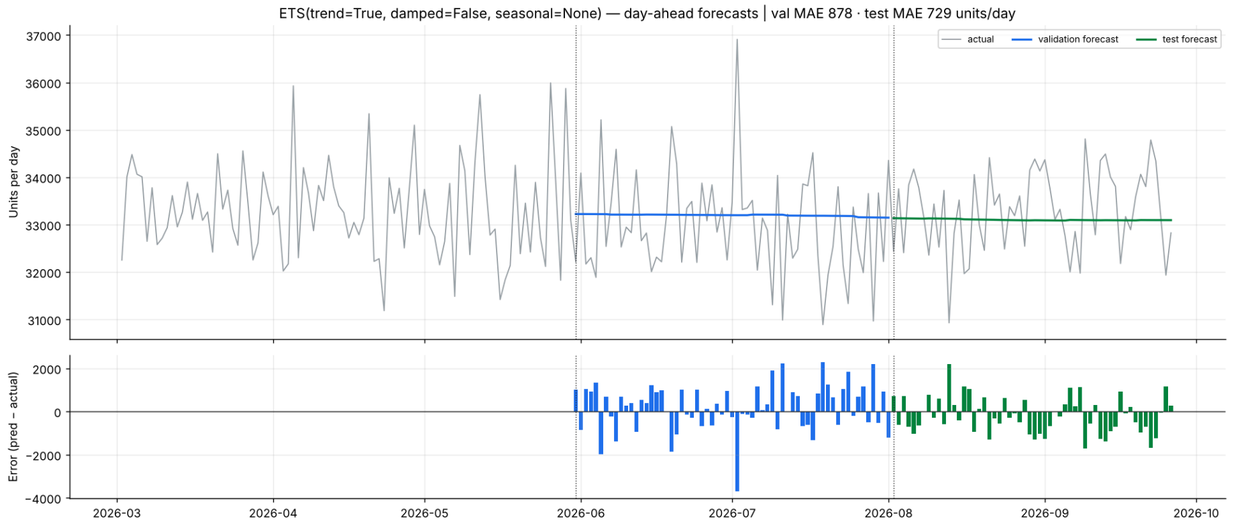
MODEL	OWNER	VAL MAE	VAL MAPE	TEST MAE	TEST RMSE	MAPE	SMAPE	MASE	BIAS	DM P
ETS ★	**benchmark**	**878**	**2.67%**	**729**	**866**	**2.19%**	**2.19%**	**0.61**	**-191**	**0.970**
ARIMAX + promo	**Rahaf**	**889**	**2.70%**	**704**	**829**	**2.12%**	**2.12%**	**0.59**	**-87**	**0.796**
Theta	**benchmark**	**892**	**2.70%**	**737**	**865**	**2.21%**	**2.22%**	**0.62**	**-170**	**0.156**
SARIMA	**Khalid**	**901**	**2.74%**	**715**	**853**	**2.15%**	**2.15%**	**0.60**	**-50**	**0.843**
XGBoost + lags	**Afnan**	**923**	**2.81%**	**710**	**838**	**2.14%**	**2.14%**	**0.59**	**-40**	**0.666**
LightGBM + lags	**benchmark**	**924**	**2.82%**	**732**	**864**	**2.21%**	**2.20%**	**0.61**	**-21**	**0.494**
28-day average	baseline	876	2.66%	732	861	2.20%	2.20%	0.61	-186	ref.
Historical mean	baseline	890	2.71%	708	843	2.13%	2.13%	0.59	-41	0.856
Seasonal naïve	baseline	1,121	3.42%	1,067	1,323	3.21%	3.22%	0.89	-99	0.000
7-day average	baseline	946	2.86%	772	942	2.33%	2.32%	0.65	-18	0.051
Naïve (yesterday)	baseline	1,291	3.90%	929	1,120	2.80%	2.80%	0.78	+27	0.000
XGBoost nowcast (same morning)	*Afnan*	*448*	*1.35%*	*371*	*496*	*1.12%*	*1.12%*	*0.31*	*-57*	*other horizon*

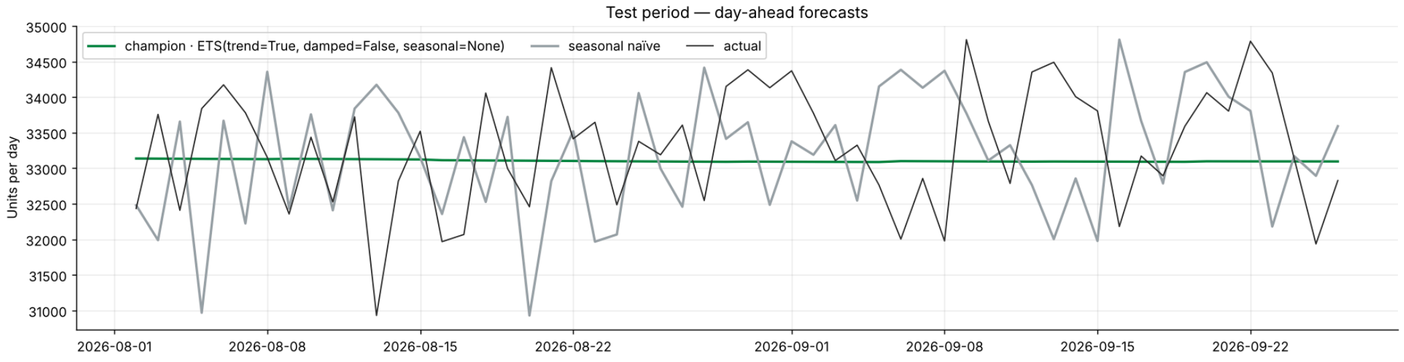
Every model under-forecasts slightly (negative bias). Ljung–Box p > 0.1 for SARIMA, ARIMAX and XGBoost: no remaining autocorrelation.

Source: results F00–F05 (*.json) · DM test, absolute loss, h = 1, HLN correction, 119 days

Forecast Backtest · ETS and All Models

Day-ahead forecasts vs actual with daily errors; every model on the test period

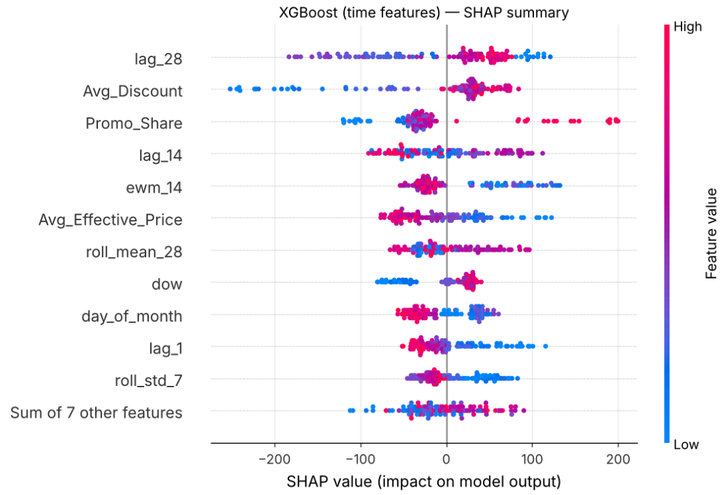


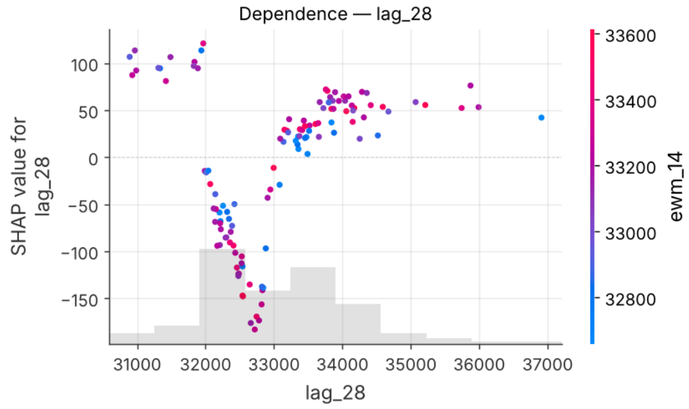


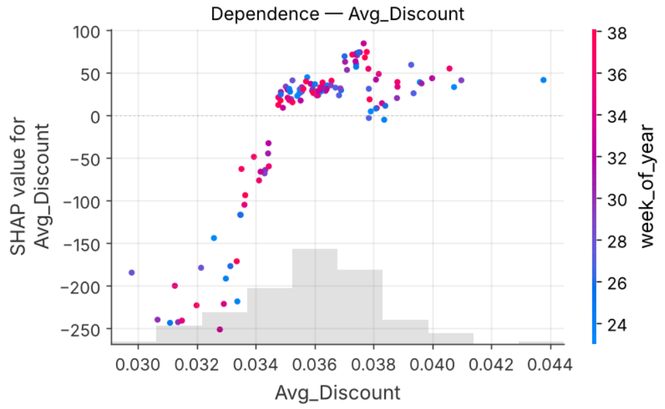
Source: figures F03a_ETS_forecast.png · 00_forecasting_test.png

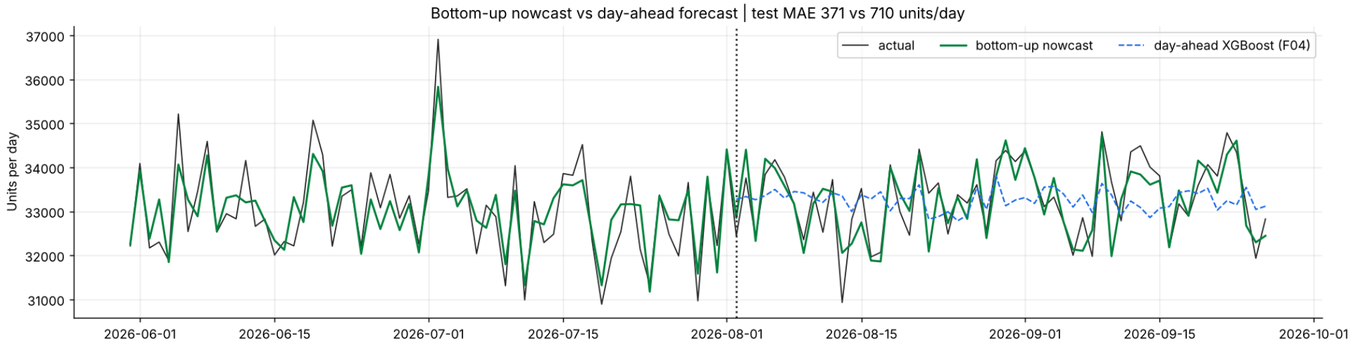
SHAP · XGBoost Forecaster & Nowcast (Afnan)

Drivers of the ML forecaster; the same-morning nowcast halves the day-ahead error









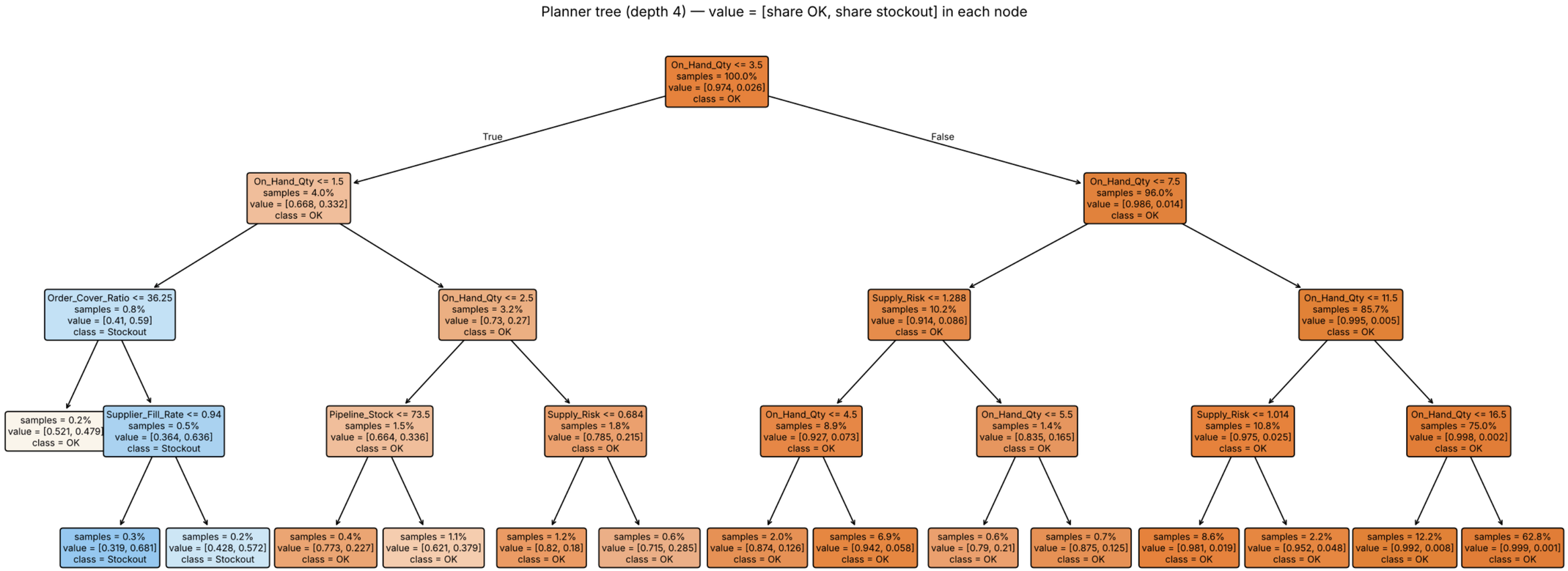
Business reading

Sales 28 and 14 days ago, average discount and promotion share drive the forecast. The nowcast adds opening stock each morning: MAE 371 vs ~710 day-ahead.

Source: figures F04_XGBoost_shap_*.png · F04b_XGBoost_nowcast.png

Interpretable Model · Decision Tree Rules (Afnan)

Top levels of the tuned tree: readable stockout rules for planners (test PR-AUC 0.347, 91% top-20% capture)



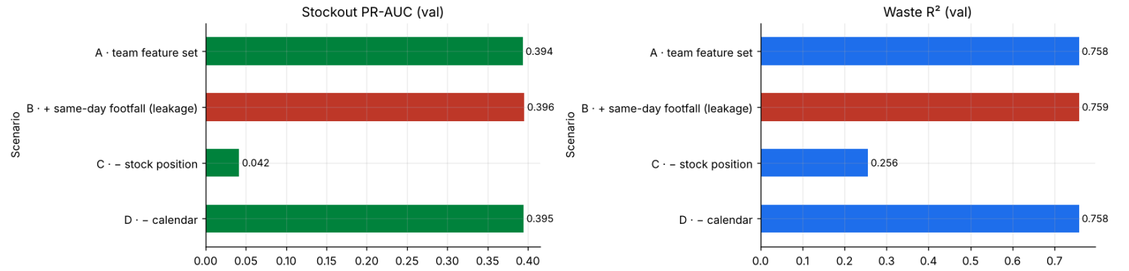
Source: figure C02_Decision_Tree_planner_tree.png

Leakage Checks & Ablation

Which columns reveal the answer, and what happens to the models when feature groups are removed

COLUMN	EVIDENCE	DECISION
On_Shelf_Availability_Pct	corr −0.88 with stockout, same day	excluded
Units_Sold / Sales_Value / Extreme_Demand	known after the day; RF F1 0.22 → 0.54	excluded
Store_Footfall	same-day visitors, only known after the day	excluded
Stockout_Flag / Relabel / TARGET_*	copies of the target	excluded
RULE_Near_Expiry	deterministic rule (shelf life ≤ 3)	excluded
On_Hand_Qty	0.6–2.4× same-day sales	kept · opening-stock assumption

FEATURE SET	FEATURES	STOCKOUT PR-AUC	ROC-AUC	WASTE R^2	WASTE MAE
A · team feature set	**23**	**0.394**	**0.946**	**0.758**	**0.513**
B · + same-day footfall (leakage)	**25**	**0.396**	**0.946**	**0.759**	**0.513**
C · − stock position	**19**	**0.042**	**0.613**	**0.256**	**0.931**
D · − calendar	**18**	**0.395**	**0.946**	**0.758**	**0.514**



Source: notebook A01_Leakage_Ablation (validation) · Phase 3 leakage review

Data Dictionary, Split & Assumptions

Feature groups used by the models (* engineered in the pipeline) and the chronological split

GROUP	COLUMNS
Inventory	On_Hand_Qty, On_Order_Qty, Order_Cover_Ratio, Pipeline_Stock*, Stock_per_Shelf_Day*
Supply	Lead_Time_Days, Supplier_Fill_Rate (+flag), Supply_Risk
Price & promotion	Regular_Price_SAR, Effective_Price, Discount_Rate, Promotion_Flag
Shelf life	Remaining_Shelf_Life_Days (+ missing flag)
Calendar	Day_Of_Week, Is_Weekend, Month, Week_Of_Year, Ramadan_Flag
Product	Category (6), Brand, Temperature_Zone (3)
Targets	TARGET_Stockout · TARGET_Waste_Qty · TARGET_Units_Sold (daily sum for forecasting)
Excluded	Record_ID, Date, Week_Start, Split, Store_Footfall (+flag), RULE_Near_Expiry

SPLIT	PERIOD	DAYS	ROWS	STOCKOUT %
Train	05 Jan 2025 – 30 May 2026	511	700,070	2.63%
Validation	31 May – 01 Aug 2026	63	86,310	2.69%
Test	02 Aug – 26 Sep 2026	56	76,716	2.75%

Assumptions

- On_Hand_Qty is opening stock, known before the day starts.
- Shelf life and supplier fill rate are known at prediction time.
- Data is synthetic: store, SKU and category attributes are random, so no location insights.

Source: PANDA_Model_Ready.csv header · Split column · Phase 3 documentation

Additional Methods, Limitations & Next Steps

Every rubric approach: what we applied, its limit, and the data that would make it stronger

APPROACH	STATUS AND LIMIT	WHAT WOULD MAKE IT STRONGER
Anomaly detection	Applied (U01): 4 methods at a 1% budget; judged by stockout / waste proxies because no labelled anomalies exist.	Labelled incidents (shrink, data errors)
Clustering	Applied (U02): 5 algorithms; segments overlap (silhouette 0.17), so used as planning groups.	Real store attributes (size, format, region)
Association rules	Applied (U03): each record treated as a basket of its conditions; no real customer baskets.	POS transactions with basket IDs
Recommendation	Applied (U04): next best action per record; no customer or product IDs for personal recommendations.	Loyalty customer × product interactions
A/B testing	Analysed (U05): promotion effect is observational; the pilot is sized but not yet run.	Randomised store pilot (treatment / control)
Store / SKU-level forecasts	~2 records per store × category × week; 20% weeks missing.	Consistent store and SKU identifiers
LSTM / deep learning	One daily series of 630 points; no pattern to learn.	Longer history, many related series

NEXT-STEP EXPERIMENTATION PLAN

Weeks 1–2
Confirm opening-stock timing and shelf-life capture with Panda

Weeks 3–4
Choose alert threshold with store managers (recall vs workload)

Weeks 5–12
A/B pilot in selected stores: waste, stockouts, service, ROI

Monthly
Retrain, monitor drift and KPI dashboard

Source: Group 10 data review · Presentation #4 rubric approach list

Code Evidence & Reproducibility

31 executed Jupyter notebooks, one shared pipeline, fixed seeds and logged results

NOTEBOOKS	CONTENT
C00–C08	Classification: baselines + 8 models (tuning, evaluation, assessment, SHAP)
R00–R07	Regression: baselines + 7 models
F00–F05	Time series: baselines + 6 models + nowcast
A01	Leakage ablation
U01–U03	Anomaly detection, clustering, association rules (targets and dates removed)
U04–U05	Recommendation (next best action) and A/B testing
E01	Explaining the black box: SHAP impact, tipping points, right / wrong cases
00_Model_Comparison	Leaderboards, bootstrap CIs, DM tests, champion decision, Excel export

Every notebook follows the same 6 steps

1 Setup → 2 Optimisation (tuned vs untuned, time-ordered CV) → 3 Evaluation (validation → test) → 4 Assessment (over-fit, robustness, latency) → 5 SHAP → 6 Business reading

PIPELINE & ENVIRONMENT

```
Pipeline([
  ("prep", ColumnTransformer([
     ("num", SimpleImputer(median) → Scaler*),
     ("cat", OneHotEncoder)])),
  ("model", Estimator)])
search = RandomizedSearchCV(pipe, space,
  cv=TimeSeriesSplit(3), scoring=metric)
# fitted on TRAIN only; refit per fold
```

pandas 3.0 · scikit-learn 1.9 · xgboost 3.4 · lightgbm 4.7 · catboost 1.2 · shap 0.54 · statsmodels 0.15 · optuna 5.0 · seed 42 · results/*.json · figures/*.png · Phase4_Model_Comparison.xlsx

Source: PANDA_Phase4_Models package (README, requirements.txt)